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# Evaluation of the microflora of public surfaces

CH5200 - Food Microbiology  
Experiment 2a



| Group 2                  |                                     |                              |
|--------------------------|-------------------------------------|------------------------------|
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**Date Submitted** : 17 May 2024

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# **1 General Introduction**

Microorganisms are prevalent on various surfaces, with varying levels of contamination depending on location and surface properties. Such microflora may contain pathogenic strains, which may cause illness (Boyce et al., 2007). Hence, surface sampling can be done to ensure that the microbial load of public surfaces is within safe limits. In this experiment, microbiological sampling and analysis was performed on a public lounge seat that is used on a regular basis to evaluate its microbial load.

## **2 Materials and Methods**

In this experiment, four sampling techniques were used: dipslide, swab, ATP, and direct contact plate methods. Sampling was conducted on four different surface categories—fingertip, body parts, a general surface, and a flat surface. For fingertip imprinting, the fingertips of each member's right hand (excluding the thumb) were directly imprinted onto individual PCA agar plates. Three body parts (palm, neck, and forearm) and three general surfaces (laboratory floor, phone case, and iPad screen) were sampled using the dipslide method. As for the flat surface, a public lounge seat located within the school was sampled using the dipslide, swab, and ATP methods.

All samples were incubated at 30°C for 24 hours. The sample colonies were counted and recorded.

### 3 Results and Discussion

Table 1 contains the microbiological counts of the PCA agar plates that were imprinted with the fingertips of each of the three group members.

**Table 1:** Microbiological status of group members' fingertips

| Members | Total CFU of 4 fingertips | Average CFU/ fingertip |
|---------|---------------------------|------------------------|
| 1       | 37                        | 9.25                   |
| 2       | 88                        | 22.00                  |
| 3       | 167                       | 41.75                  |

As shown in Table 1, the average levels of CFU/fingertip of each member varied, with member 3 displaying a higher microorganism count. This variation in CFU count was expected. Due to a complex interaction between internal and external factors, each person's skin microbiome composition differs greatly (Khan & Koh, 2024). Differences in CFU counts amongst each member likely reflect differing surface contact prior to fingertip imprinting and varying personal hygiene habits.

Next, the dipslide method was used to sample various surfaces. Tables 2 and 3 contain the results interpreted using a model density chart (Annex Figure B1 and Table B2 compared to Table C1).

**Table 2:** Microbiological status of group members' body parts obtained via dipslide

| Members | Body part | CFU/ 25cm <sup>2</sup> |
|---------|-----------|------------------------|
| 1       | Forearm   | 25                     |
| 2       | Neck      | 125                    |
| 3       | Palm      | 25                     |

**Table 3:** Microbiological status of general surfaces obtained via dipslide

| Members | General surface  | CFU/ 25cm <sup>2</sup> |
|---------|------------------|------------------------|
| 1       | Ipad screen      | 25                     |
| 2       | Phone case       | 25                     |
| 3       | Laboratory floor | 125                    |



From Table 2, it can be inferred that the neck region harbored higher levels of microorganisms as compared to the wrist and palm regions. This may be attributed to the warmer and more humid environment at the neck region, which favours microbial growth. The neck is also less exposed to mechanical abrasion compared to the wrist and palm, which are more frequently cleaned or come into contact with surfaces that remove surface bacteria. Additionally, the neck is closer to mucosal surfaces, which may serve as microbial transfer sources.

Table 3 shows that the laboratory floor has a higher CFU count than the surfaces of the iPad and phone case. The floor is subjected to constant foot traffic, introducing and redistributing microorganisms from various sources. Furthermore, floors frequently gather organic matter, dust, and dirt, providing nutrients for bacterial growth. While personal devices are regularly cleaned due to their visibility and frequent handling, floors may not receive the same level of attention, over time allowing microbial growth and spread. Porous or textured floor surfaces may also harbor bacteria more effectively than smooth, non-porous surfaces of electronic devices.

Thereafter, the public lounge seat located within the school was sampled using 3 methods, with the results summarized in Table 4. Hygiene scores were also assigned according to a standard used by food factories (Annex Figure B1 and Table B2 compared to Table D1).

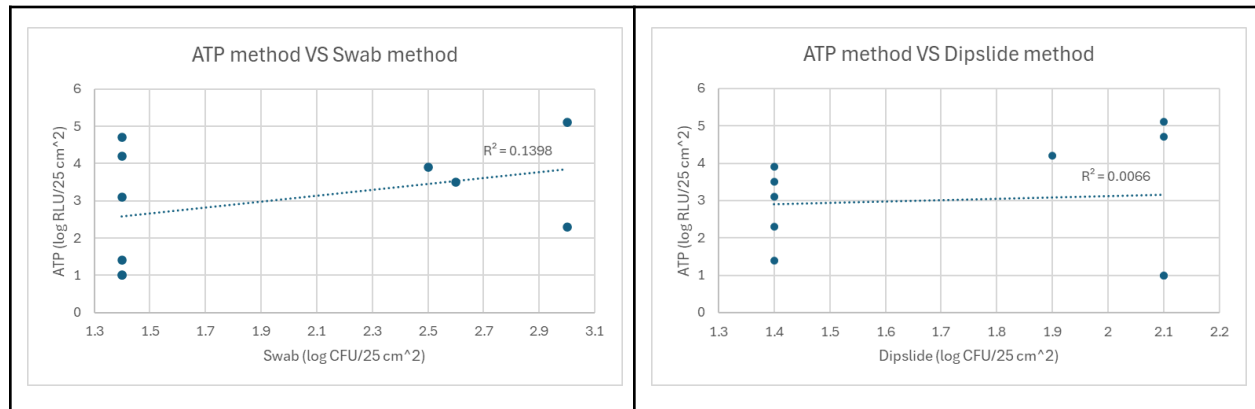
**Table 4:** Microbiological status of lounge seat surface in SBS Lobby, NTU

| Method   | CFU/ 25cm <sup>2</sup> | Hygiene score | Hygiene status |
|----------|------------------------|---------------|----------------|
| Dipslide | 25                     | 2             | Satisfactory   |
| Swab     | 25                     | 2             | Satisfactory   |
| Method   | RLU/ 25cm <sup>2</sup> | Hygiene score | Hygiene status |
| ATP      | 1356                   | -             | Fail           |

Both the dipslide and swab methods yielded a hygiene score of 2, reflecting that while basic cleanliness standards are met, more effort could be spared for improvement. On the other hand, the ATP test failed the cleanliness assessment, suggesting the presence of residual organic matter or microbial contamination that is undetectable through culture-based methods (i.e. non-culturable microorganisms). ATP testing measures adenosine triphosphate, an essential molecule found in all living cells. It provides a rapid indication of overall surface contamination. The ATP test's sensitivity to organic residues or non-culturable microorganisms signifies insufficient cleaning, which

may be the cause of the discrepancy between the ATP results and the culture-based techniques. This emphasizes how crucial it is to use a variety of hygiene assessment techniques in order to get a thorough grasp of surface cleanliness and to spot possible areas that need more stringent sanitation regulations.

The oversensitivity of the ATP test towards non-microbial matter may render it unreliable occasionally when detecting microbial contamination. This unreliability can be observed from the results of all the groups (Annex E1).



**Figure 1:** [left] ATP (log RLU/ 25cm<sup>2</sup>) against the Swab method (log CFU/ 25cm<sup>2</sup>)  
[right] ATP (log RLU/ 25cm<sup>2</sup>) against the Dipslide method (log CFU/ 25cm<sup>2</sup>)

Based on Figure 1, the  $R^2$  values (0.1398 and 0.0066) are close to 0, indicating little to no correlation between the dipslide/swab results and the ATP results, indicating that ATP might not be a suitable test for determining the microbiological status of a surface.

#### Question from the manual:

##### ***Is the ATP method suitable for quantitative measurements? Explain.***

ATP (adenosine triphosphate) testing offers rapid quantitative assessments of surface cleanliness by detecting ATP levels via bioluminescence, with results expressed in Relative Light Units (RLUs). Although this approach gives relatively quick feedback on the existence of organic material, it is not very accurate at detecting microbial contamination. This restriction results from the fact that ATP tests identify all ATP sources, including non-microbial residues such as food particles or human cells, leading to overestimations of microbial presence. Studies have shown that there is a weak correlation between RLU values and aerobic colony count, with an R value of only 0.244 (Van Arkel et al., 2021). This study suggests that ATP may not be entirely reliable for assessing microbial contamination levels. Hence, while ATP testing provides prompt general cleanliness assessment, it should be complemented with culture-based methods to accurately evaluate and quantify microbial contamination.

## 4 Conclusions and Recommendations

This experiment assessed microbial contamination across various surfaces using direct contact plate, dipslide, swab, and ATP bioluminescence methods. Fingertip microbial variation among individuals reflected differences in the skin microbiome, recent contact, and hygiene habits. Culture-based methods revealed moderate microbial presence, with higher CFU counts on the neck and laboratory floor, which are areas that provide a more favorable environment for microbial growth due to exposure to organic matter, warmth, and moisture. While the lounge seat showed satisfactory results in culture-based assessments, it failed the ATP testing, highlighting the limitation of relying solely on ATP measurements for evaluating microbial contamination.

Based on the findings of these experiments, it is recommended to implement a multi-method approach to evaluate surface cleanliness, as no singular method could provide a fully comprehensive assessment. ATP testing rapidly provides results, but it is accompanied by the drawback of producing false positives due to its high sensitivity to all organic residues. Culture-based methods, which are better at reflecting viable microbial loads, should be implemented as a complement to more accurately assess microbial contamination.

Regular cleaning and sanitation, especially of high-contact or high-exposure areas like floors and shared seating, should be maintained or improved. Additionally, public awareness regarding hand hygiene remains crucial, as microorganisms are constantly present on human skin.

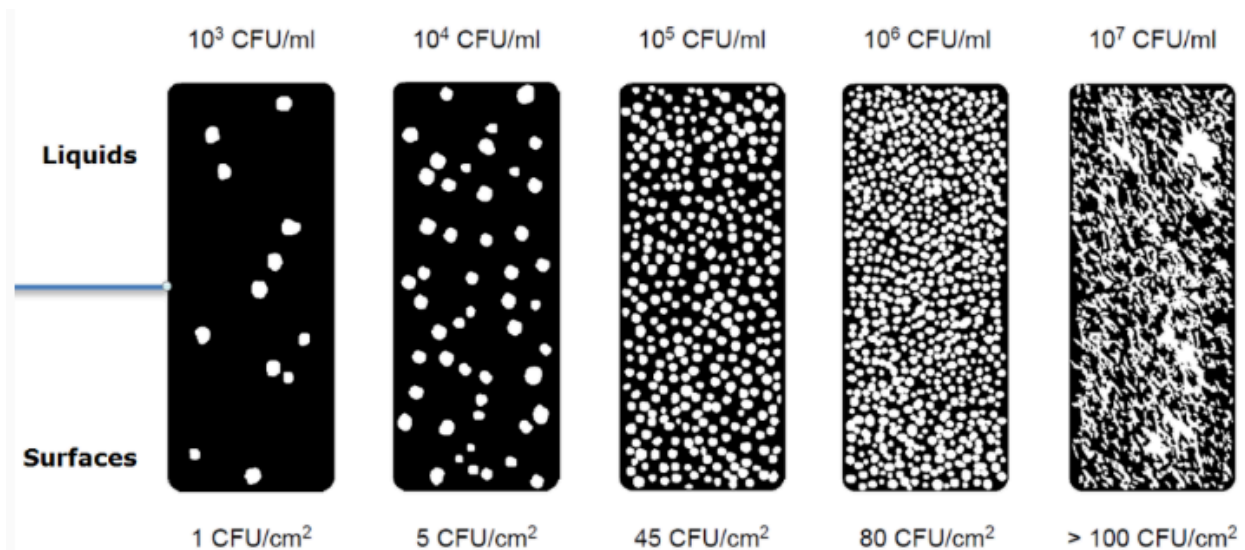
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<https://doi.org/10.4167/jbv.2024.54.3.191>
- Van Arkel, A., Willemsen, I., & Kluytmans, J. (2021). The correlation between ATP measurement and microbial contamination of inanimate surfaces. *Research Square (Research Square)*. <https://doi.org/10.21203/rs.3.rs-428499/v1>

## Annex

**Table A1:** Microbiological status (average CFU calculations) of group members' fingertips

| Members | Total CFU of 4 fingertips | Average CFU/ fingertip  |
|---------|---------------------------|-------------------------|
| 1       | 37                        | $\frac{37}{4} = 9.25$   |
| 2       | 88                        | $\frac{88}{4} = 22.00$  |
| 3       | 167                       | $\frac{167}{4} = 41.75$ |

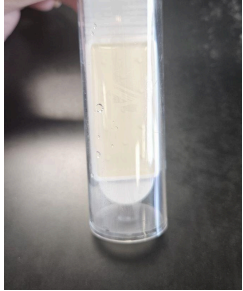


**Figure B1:** Model density chart used to determine microbial count, by comparing the microbial density on the dipslide with the chart

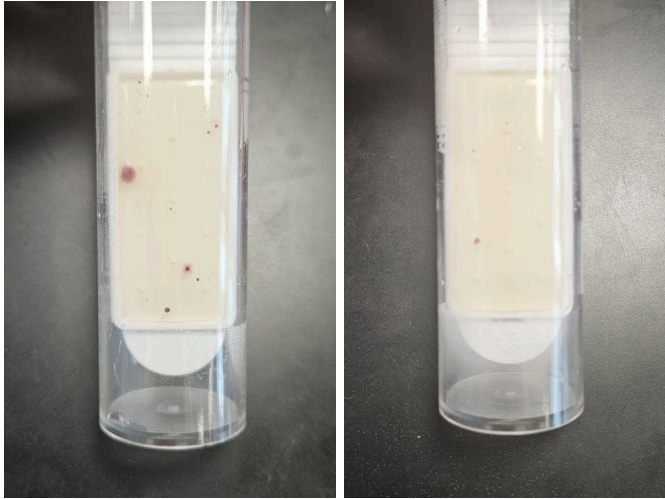
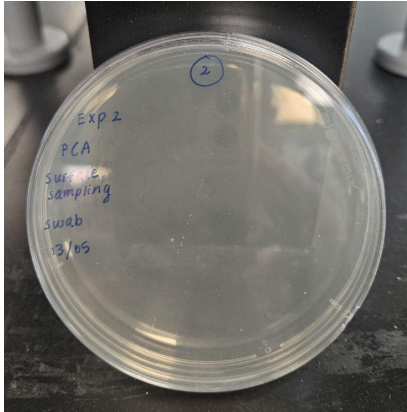
**Table B2:** Hygiene score standard for food factory

| Number of CFU on 25cm <sup>2</sup> | Score | Remark         |
|------------------------------------|-------|----------------|
| 0-2                                | 0     | Very Good      |
| 3-9                                | 1     | Good           |
| 10-29                              | 2     | Satisfactory   |
| 30-90                              | 3     | Questionable   |
| >90                                | 4     | Unsatisfactory |

**Table C1:** Microbiological status of group members' body parts and general surfaces obtained via dipslide method

| Members | Body part | Dipslide for body part                                                              | CFU/ 25 cm <sup>2</sup> | General surface  | Dipslide for general surface                                                          | CFU/ 25 cm <sup>2</sup> |
|---------|-----------|-------------------------------------------------------------------------------------|-------------------------|------------------|---------------------------------------------------------------------------------------|-------------------------|
| 1       | Forearm   |    | 25                      | Ipad screen      |    | 25                      |
| 2       | Neck      |   | 125                     | Phone case       |   | 25                      |
| 3       | Palm      |  | 25                      | Laboratory floor |  | 125                     |

**Table D1:** Microbiological status of lounge seat surface in SBS Lobby, NTU

| Method                  | Results                                                                                                                                      | CFU/ 25cm <sup>2</sup> |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Via dipslide            |  <p>*Both sides of the dipslide were used for sampling</p> | 25                     |
| Via swab                |                                                            | 25                     |
| Method                  | Results                                                                                                                                      | RLU/ 25cm <sup>2</sup> |
| Via ATP bioluminescence | -                                                                                                                                            | 1356                   |

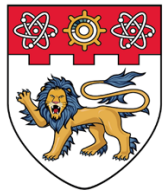
**Table E1:** Results for surface sampling from all groups

|                          | CFU/25 cm <sup>2</sup>       |             | log CFU/25 cm <sup>2</sup> |             | RLU/25 cm <sup>2</sup><br>(Pass/ Fail) | log RLU/25 cm <sup>2</sup> |
|--------------------------|------------------------------|-------------|----------------------------|-------------|----------------------------------------|----------------------------|
| Surface                  | Via<br>dipslide              | Via<br>swab | Via<br>dipslide            | Via<br>swab | Via ATP                                | Via ATP                    |
| Lab table                | 25                           | 25          | 1.4                        | 1.4         | 25 (pass)                              | 1.4                        |
| SBS<br>Lobby<br>couch    | 25                           | 25          | 1.4                        | 1.4         | 1356 (fail)                            | 3.1                        |
| Lift button              | 125                          | <25         | 2.1                        | <1.4*       | 0** (pass)                             | 0                          |
| Water<br>cooler<br>drain | 125                          | 1120        | 2.1                        | 3.0         | 129648 (fail)                          | 5.1                        |
| Toilet seat              | 125                          | <25         | 2.1                        | <1.4*       | 0** (pass)                             | 0                          |
| Laboratory<br>sink 1     | 25 - 125<br>(75 was<br>used) | <25         | 1.9                        | <1.4*       | 17291 (fail)                           | 4.2                        |
| Toilet rim               | 25                           | 350         | 1.4                        | 2.5         | 8141 (fail)                            | 3.9                        |
| Laboratory<br>sink 2     | <25                          | 900         | <1.4*                      | 3.0         | 214 (fail)                             | 2.3                        |
| Laboratory<br>sink 3     | <25                          | 430         | <1.4*                      | 2.6         | 3396 (fail)                            | 3.5                        |
| Toilet door<br>handle    | 125                          | <25         | 2.1                        | <1.4*       | 49586 (fail)                           | 4.7                        |

\*It is important to note that microbial counts of <1.4 log CFU/25 cm<sup>2</sup> were taken as 1.4 when plotting the graph.

\*\*Value of 1 was used when calculating log RLU/25 cm<sup>2</sup> to prevent math error.





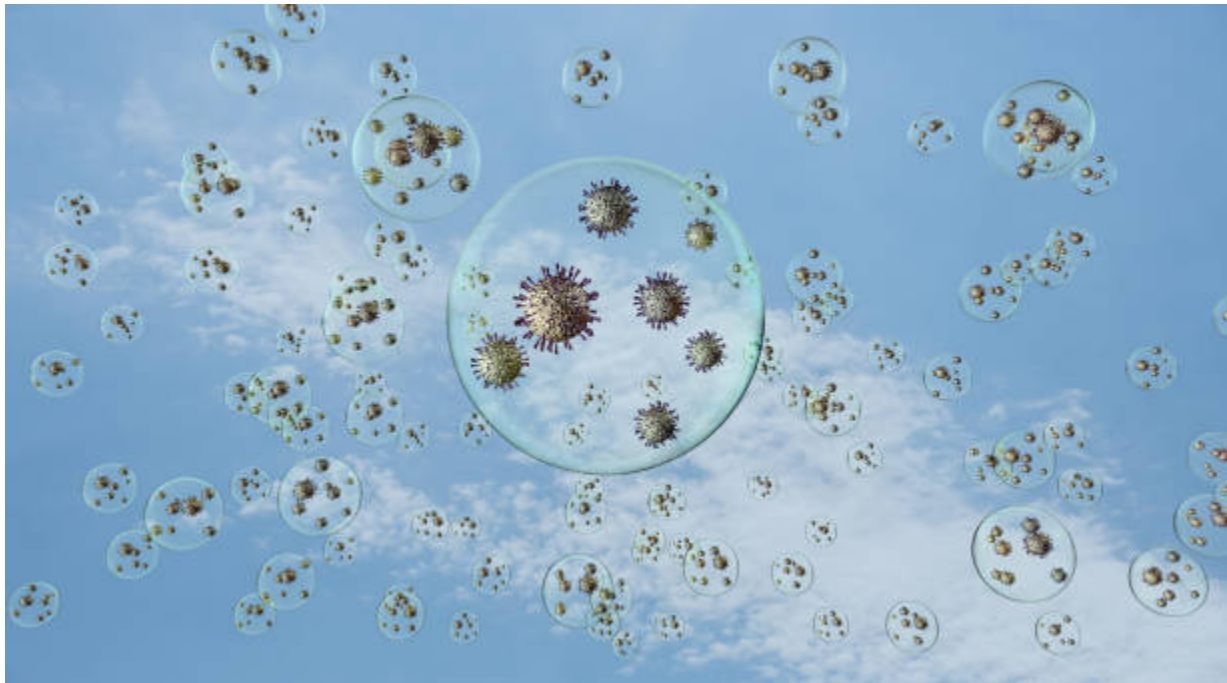
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# Sampling and evaluation of airborne microflora

CH5200 - Food Microbiology  
Experiment 2b



| Group 2                  |                                     |                              |
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**Date Submitted :** 21 May 2025

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# **1 General Introduction**

Compared to soil, food, or even water, air is a harsh and hostile environment to microbial growth due to the effects of radiation and desiccation (Adams et al., 2024). However, microbes are still able to disperse through airborne routes, contaminating any suitable, exposed surface for growth. In this experiment, different air sampling instruments were utilized and compared to evaluate the microbiological status of public airspaces, assessing the risk presented by airborne microflora as well as evaluating the performance of the different instruments.

## **2 Materials and Methods**

In this experiment, three air sampling instruments were used: MAS-100 NT® and Hycon RCS Plus, with each apparatus sampling 100 L of air, and Emtex P100 sampling 200 L of air. Sampling was conducted in Classroom 1 (CR1) lecture theater on Level 1 in the School of Biological Sciences at NTU. The airspace had a noticeably musty odor and was an enclosed area with little ventilation. The agar plates/ strips from the various air sampling instruments were then incubated until a suitable amount of growth was observed, and the number of colonies was examined. Further isolation and characterization of pure bacterial colonies (excluding fungi) were done, and an evaluation of the bacterial diversity was made.

### 3 Results and Discussion

Air sampling was performed inside Classroom 1 (CR1) of the SBS Building using various instruments, namely the Emtek P100, MAS, and Hycon. The colony count and concentration recorded from the air sampling are recorded in Table 1.

**Table 1:** Colony count and microorganism concentration of CR1 located indoors within the SBS Building, determined using Emtek P100, MAS and Hycon air samplers.

| Air sampling method<br>(Instrument used) | Colony count/<br>$\log_{10}\text{CFU}$ | Concentration of<br>microorganisms/ $\log_{10}\text{CFU/m}^3$ |
|------------------------------------------|----------------------------------------|---------------------------------------------------------------|
| <b>Emtek P100 (Group 9)</b>              | 16                                     | 1.9                                                           |
| <b>MAS (Group 4)</b>                     | 14                                     | 2.1                                                           |
| <b>Hycon (Group 2)</b>                   | Not countable                          | Not countable                                                 |

From Table 1, it can be inferred that the MAS instrument yielded a higher concentration of microorganisms in the air sampled, with Emtek P100 yielding a slightly lower concentration (the difference is insignificant,  $<0.5 \log_{10}\text{CFU/m}^3$  difference). On the other hand, Hycon did not yield a countable result due to condensation smear on the agar strip (Annex Figure A2), resulting in a sample with little/ no distinct colonies.

Colonies on the agar plates/ strips were then streaked on separate agar plates for further identification. The identified bacterial groups are recorded in Table 2.

**Table 2:** Microorganism species identified in CR1 SBS Building using Emtek P100, MAS and Hycon air samplers.

| Emtek P100 (Group 9)           | MAS (Group 4)           | Hycon (Group 2)            |
|--------------------------------|-------------------------|----------------------------|
| <i>Bacillus spp.</i>           |                         |                            |
| <i>Enterobacteriaceae spp.</i> |                         | -                          |
| <i>Staphylococcus spp.</i>     | -                       | <i>Staphylococcus spp.</i> |
| -                              | <i>Pseudomonas spp.</i> | -                          |
| Fungi                          |                         |                            |

*The subdivision of microorganism in groups is located in the annex (Annex Table B1)*

### **3.1. Common microorganisms found in air**

In CR1 (our sampling location), the presence of *Bacillus spp.* and fungi were detected by all three instruments. The air is mainly dominated by Gram-positive rod-shaped bacteria (e.g., *Bacillus spp.*). Their thick cell wall structure—mainly composed of peptidoglycan—acts as a protective barrier, allowing for increased viability. Additionally, both *Bacillus spp.* and fungi are capable of producing spores that can survive in the air, further supporting their survivability and persistence in the air. *Enterobacteriaceae spp.* and *Staphylococcus spp.* were also detected and attributed to human origin, as CR1 is a lecture theater where human occupancy is common. Another contributor to the diversity of the airborne microflora in CR1 is the carpet flooring, which is not frequently cleaned and thus may accumulate organic waste, including dirt and dust from the constant foot traffic.

Comparing with the species analysis of other locations (Annex Table B1), similar microbial diversity (excluding fungi, as the experiment excludes fungal identification) was observed, with *Bacillus spp.* and fungi being ubiquitous and other strains including *Pseudomonas spp.*, *Staphylococcus spp.*, and *Enterobacteriaceae spp.* also being present. Noticeably, *Lactobacillus spp.* were found in the Clade Room, which is a probiotic that is commonly traced to human origin, which aligns with the Clade Room being a communal lounge that hosts students frequently (Liu et al., 2022).

### **3.2. Comparison between indoor and outdoor air quality**

Compared with the microbial yield of other locations (Annex Table A1), both indoor and outdoor environments have relatively similar microbial loads (i.e.,  $<0.5 \log_{10} \text{CFU/m}^3$  difference). While microbial loads show no major differences, the difference in microflora diversity was distinct (focusing on bacteria diversity, excluding fungi), with Gram-negative bacteria observed mostly in indoor environments due to increased susceptibility to outdoor environmental stressors like desiccation and UV radiation. In particular, *Staphylococcus spp.* were only observed in indoor environments. *Staphylococcus spp.* are commonly found on human skin, in the nose, and in mucous membranes. Thus, their presence is attributed to a heightened human presence in indoor locations.

A relatively unexpected result was the presence of *Enterobacteriaceae spp.* in the air sampled at SBS Building Level 2 outdoor area. *Enterobacteriaceae spp.* are Gram-negative bacteria that are susceptible to desiccation, making them less common in open-air environments, albeit not completely impossible. The detection of *Enterobacteriaceae spp.* could be attributed to the abundance of vegetation and soil in

the surrounding area of the outdoor area, or the presence of animal (including birds and rodents) waste, all of which are potential sources of *Enterobacteriaceae spp.*

### **3.3. Location with the highest yield**

The highest yield ( $2.6 \log_{10}\text{CFU}/\text{m}^3$ ) was unexpectedly found at LKC Bridge, which is an open-air environment. This may be due to a recently produced aerosol from a human or animal source (Adams et al., 2024). Another possibility is human error or random fluctuation. The high yield cannot be validated as sampling was only conducted by one group, without additional replicate results. In general, the microbial load across the different locations ranged from 1.8 to  $2.6 \log_{10}\text{CFU}/\text{m}^3$ , which is within the WHO's recommended limit for airborne microbial levels at  $3.0 \log_{10}\text{CFU}/\text{m}^3$  (Heseltine & Rosen, 2009).

### **3.4. Comparison of the three instruments**

The Emtex P100 uses a vacuum pump attached with a remote head for direct impaction onto an agar plate, highlighting its suitability for high-grade cleanroom settings that require remote air sampling. MAS operates through direct impaction on an agar plate via a perforated head and is well suited for routine monitoring in controlled environments. Hycon uses centrifugal impaction onto an agar strip and is best suited for general environmental monitoring where simplicity, portability, and ease of use are preferred over advanced configurability.

While the microbial load results of all groups (Annex Table A1) does indicate a slight difference in yield, it cannot be concluded that any instrument has a higher yield compared to the others, as the differences observed were within  $0.5 \log_{10}\text{CFU}/\text{m}^3$  of each other (results are only significant at  $>0.5 \log_{10}\text{CFU}/\text{m}^3$  difference). Additionally, no replicates were performed, hence variations in microbial yield may be associated with random environmental fluctuations. Hence, it can be concluded that the three instruments were of similar effectiveness and accuracy compared to each other. However, this should be further validated by an additional measurement involving sufficient replicates.

## 4 Conclusions and Recommendations

In conclusion, an evaluation of the air quality in various locations revealed that the majority of the airborne microflora is made up of fungi and *Bacillus spp.*, with Gram-negative bacteria being more prevalent indoors. The microbial loads across the different locations were assessed to be of acceptable quality according to WHO standards and thus assessed to be of acceptable risk (Heseltine & Rosen, 2009).

The performances of the three selected instruments did not show significant differences with each other, which suggests that all three are acceptable for use without any major biases. However, the Hycon was noticeably more difficult to interpret, as the agar strips were prone to condensation smear, which happened to two groups out of four. Hence, extra caution should be exercised when using the Hycon. Furthermore, replicate measurements should be performed to reduce the impact of random fluctuation.

Emtek P100, MAS, and Hycon are all recognized tools for microbial air monitoring, and selecting the most appropriate instrument should be guided by environmental requirements, regulatory considerations, and operational workflow.

## References

Adams, M. R., McClure, P. J., & Moss, M. O. (2024). *Food microbiology*. Royal society of chemistry.

Heseltine, E., & Rosen, J. (Eds.). (2009). WHO guidelines for indoor air quality: dampness and mould.

Liu, M., Gan, Z., Shen, B., Liu, L., Zeng, W., Li, Q., & Liu, H. (2022). Occupants contribute to pathogens and probiotics in indoor environments. *Building and Environment*, 213, 108863.



## Annex

**Table A1:** Colony count and microorganism concentration of the four chosen locations (comparison across all groups) located within the SBS Building, determined using Emtex P100, MAS and Hycon air samplers

| Location                                                                | Emtek P100/<br>$\log_{10}\text{CFU/m}^3$<br>(Flow rate:<br>200 L/min) | MAS/<br>$\log_{10}\text{CFU/m}^3$<br>(Flow rate: 100<br>L/min) | Hycon/<br>$\log_{10}\text{CFU/m}^3$<br>(Flow rate: 100<br>L/min) | Average<br>$\log_{10}\text{CFU/m}^3$               |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------|
| <b>Groups 1, 5, 7:</b><br>Open Air Level 2<br>SBS Building<br>(Outdoor) | CFU count: 26<br>$\log_{10}(26/0.2) =$<br><u><b>2.1</b></u>           | CFU count: 17<br>$\log_{10}(17/0.1)$<br><u><b>= 2.2</b></u>    | CFU count: 7<br>$\log_{10}(7/0.1)$<br><u><b>= 1.8</b></u>        | $\frac{2.1 + 2.2 + 1.8}{3}$<br><u><b>= 2.0</b></u> |
| <b>Group 3:</b><br>LKC Bridge<br>(Outdoor)                              | Not done                                                              | CFU count: 44<br>$\log_{10}(44/0.1)$<br><u><b>= 2.6</b></u>    | Not countable                                                    | <u><b>2.6</b></u>                                  |
| <b>Groups 2, 4, 9:</b><br>CR1<br>SBS Building<br>(Indoor)               | CFU count: 16<br>$\log_{10}(16/0.2)$<br><u><b>= 1.9</b></u>           | CFU count: 14<br>$\log_{10}(14/0.1)$<br><u><b>= 2.1</b></u>    | Not countable                                                    | $\frac{1.9 + 2.1}{2}$<br><u><b>= 2.0</b></u>       |
| <b>Groups 6, 8, 10:</b><br>Clade room<br>SBS Building<br>(Indoor)       | CFU count: 35<br>$\log_{10}(35/0.2)$<br><u><b>= 2.2</b></u>           | CFU count: 9<br>$\log_{10}(9/0.1)$<br><u><b>= 2.0</b></u>      | CFU count: 15<br>$\log_{10}(15/0.1)$<br><u><b>= 2.2</b></u>      | $\frac{2.2 + 2.0 + 2.2}{3}$<br><u><b>= 2.1</b></u> |

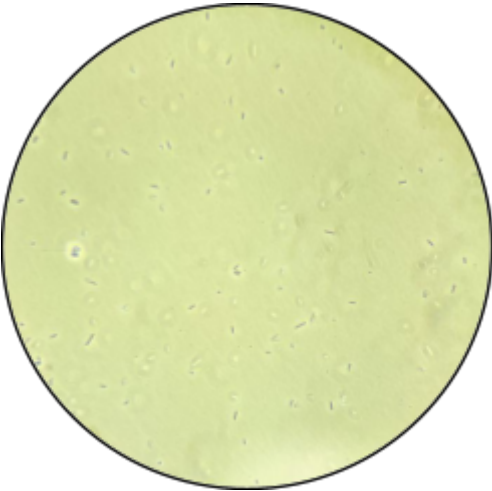
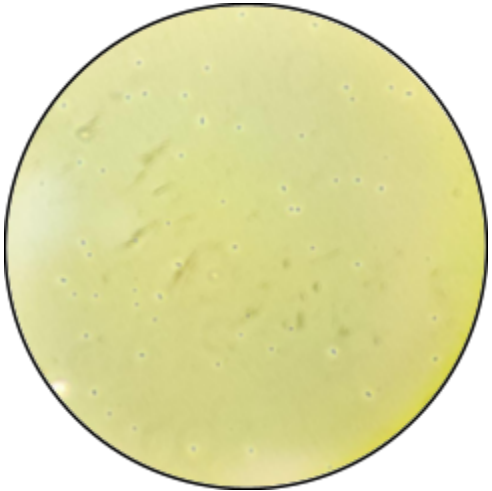
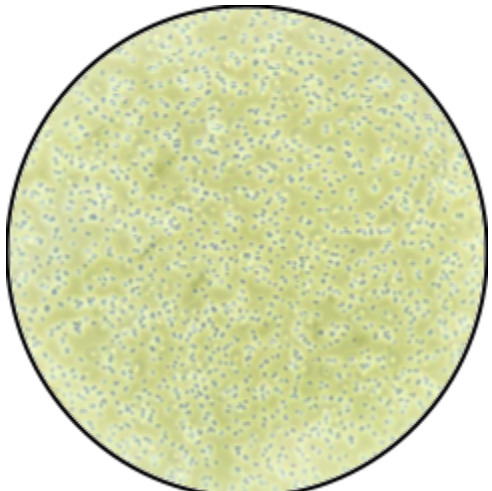
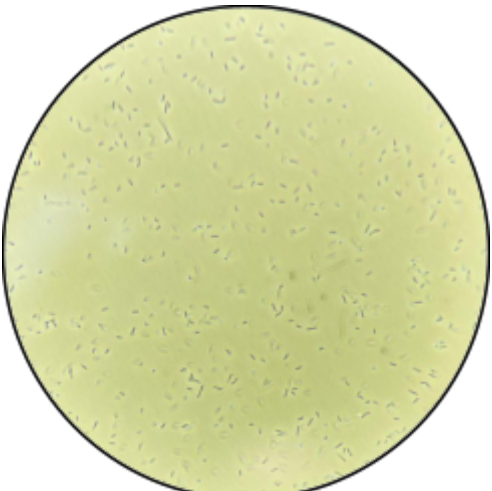


**Figure A2:** Image of air sample colonies on agar strip used for Hycon instrument (CR1 SBS Building), indicating the presence of condensation smear resulting in non-countable results

**Table B1:** Microorganism species identified in the four chosen locations (comparison across all groups) located within the SBS Building, determined using Emtek P100, MAS and Hycon air samplers

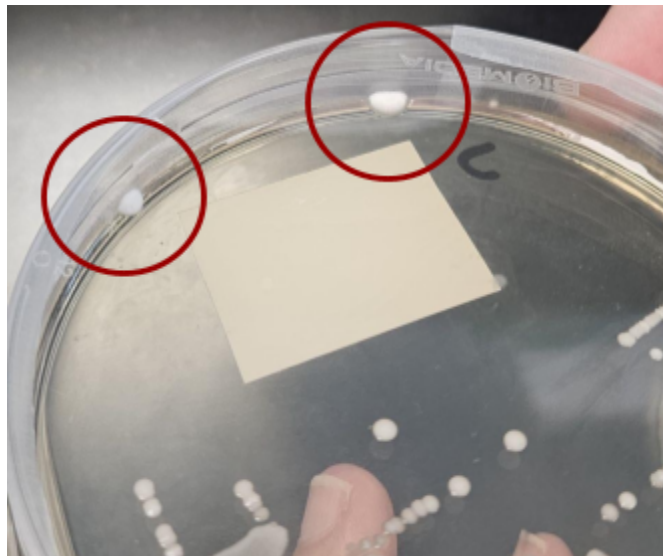
| Location                                                   | Indoor/<br>Outdoor | Emtek P100                     | MAS                            | Hycon                      |
|------------------------------------------------------------|--------------------|--------------------------------|--------------------------------|----------------------------|
| <b>Groups 1, 5, 7:</b><br>Open Air Level 2<br>SBS Building | Outdoor            | <i>Bacillus spp.</i>           |                                |                            |
|                                                            |                    | -                              | <i>Enterobacteriaceae spp.</i> | -                          |
|                                                            |                    | Fungi                          | -                              |                            |
| <b>Group 3:</b><br>LKC Bridge                              | Outdoor            | -                              | <i>Bacillus spp.</i>           | -                          |
|                                                            |                    | -                              | Other unknown species present  | -                          |
| <b>Groups 2, 4, 9:</b><br>CR1<br>SBS Building              | Indoor             | <i>Bacillus spp.</i>           |                                |                            |
|                                                            |                    | <i>Enterobacteriaceae spp.</i> |                                | -                          |
|                                                            |                    | <i>Staphylococcus spp.</i>     | -                              | <i>Staphylococcus spp.</i> |
|                                                            |                    | -                              | <i>Pseudomonas spp.</i>        | -                          |
|                                                            |                    | Fungi                          |                                |                            |
| <b>Groups 6, 8, 10:</b><br>Clade Room<br>SBS Building      | Indoor             | <i>Bacillus spp.</i>           |                                |                            |
|                                                            |                    | <i>Enterobacteriaceae spp.</i> |                                | -                          |
|                                                            |                    | <i>Staphylococcus spp.</i>     |                                |                            |
|                                                            |                    | -                              | <i>Lactobacillus spp.</i>      |                            |
|                                                            |                    | Fungi and Yeast                | -                              |                            |

**Table B2:** Image of four colonies sampled and isolated from the agar strip used for Hycon instrument (CR1 SBS Building), streaked and cultured on NA (nutrient agar) plates. The colonies were viewed under phase contrast at 1000x magnification (using 100x objective lens with immersion oil)

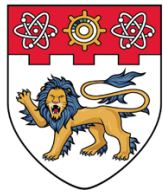
| Colony A                                                                            | Colony B                                                                             |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|    |    |
| Colony C                                                                            | Colony D                                                                             |
|  |  |

**Table B3:** Identification of the four colonies sampled and isolated from the agar strip used for Hycon instrument (CR1 SBS Building)

| Colony   | Shape  | Motility   | KOH                  | Amino peptidase      | Oxidase | Catalase | Group                       |
|----------|--------|------------|----------------------|----------------------|---------|----------|-----------------------------|
| <b>A</b> | Rod    | Motile     | -<br>(Gram positive) | -<br>(Gram positive) | -       | +        | <i>Bacillus spp.</i>        |
| <b>B</b> | Coccus | Non motile | -<br>(Gram positive) | -<br>(Gram positive) | -       | +        | <i>Staphylo-coccus spp.</i> |
| <b>C</b> | Coccus | Non motile | -<br>(Gram positive) | -<br>(Gram positive) | -       | +        | <i>Staphylo-coccus spp.</i> |
| <b>D</b> | Rod    | Motile     | -<br>(Gram positive) | -<br>(Gram positive) | -       | +        | <i>Bacillus spp.</i>        |



**Figure B4:** Image of mould colonies found on the NA plate of Colony C, indicating the presence of mould in CR1 SBS Building (using Hycon instrument)



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# Evaluation of the microbiological quality of food products

CH5200 - Food Microbiology  
Experiment 3



| Group 2                  |                                     |                              |
|--------------------------|-------------------------------------|------------------------------|
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**Date Submitted** : 23 May 2025

**Supervisors** : Dr. Sze Chun Chau and Dr. Wilma Hazeleger

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# 1 General Introduction

Microorganisms, both beneficial and harmful, are commonly found in food products depending on various intrinsic factors (e.g., pH, water activity, nutritional content, and presence of antimicrobial compounds) and extrinsic factors (e.g., temperature, relative humidity, and composition of gases in the storage environment). Microbial contamination analysis is carried out to verify that food products remain within acceptable microbial limits, especially those susceptible to pathogenic contamination. This experiment aims to assess microbial load on a range of food products, namely fermented products, dried spices and herbs, and fresh products.

## 2 Materials and Methods

In this experiment, three food products were evaluated, namely five-spice powder (in a bottle), natto (styrofoam-packed), and enoki mushrooms (plastic-wrapped). All materials procured were store-bought, within their prescribed expiry dates, and visually evaluated to be without any serious defects or obvious spoilage. In particular, the natto and enoki mushrooms were kept in refrigeration throughout storage. Total viable counts (TVC) using PCA and fungal counts using DRBC were performed for the samples. Additional investigations for the presence of spore-forming bacteria, lactic acid bacteria, *C. perfringens*, and *Enterobacteriaceae spp.* were performed on relevant food products as well.

### 3 Results and Discussion

Five-spice powder, natto, and enoki mushrooms were processed using a stomacher. The produced homogenates were subjected to various microbial tests, which can be seen in Table 1.

**Table 1:** Recorded results of various microbial testing performed on five-spice powder, natto, and enoki mushrooms

| Samples/ Tests                 | Five-spice powder | Natto         | Enoki mushroom |
|--------------------------------|-------------------|---------------|----------------|
| Total Viable Count (TVC)       | 3.4               | >8.8          | 8.4            |
| Spore Forming bacteria (Spf)   | 2.8               | 4.8           | Not performed  |
| Yeast and Mold                 | <2.3              | <2.3          | 5.3            |
| <i>Clostridium perfringens</i> | <0.3              | Not performed | Not performed  |
| Lactic Acid Bacteria (LAB)     | Not performed     | <1.0          | 4.5            |
| <i>Enterobacteriaceae</i>      | Not performed     | Not performed | <3.0           |

#### 3.1. Herbs and spices (Five-spice powder)

Five-spice powder is a common seasoning used in Chinese cuisine and is usually composed of a blend of star anise, cloves, Chinese cinnamon, Sichuan peppercorns, and fennel seeds, in a dried powdered form. From Table 1, five-spice powder displayed a high sterility for all tests conducted, being under the lower threshold for detection for the spore former, yeasts, molds and *C. perfringens* tests. The TVC and spore former count (3.4 and 2.8 log<sub>10</sub>CFU/g) were both moderately high, suggesting that the microflora of five-spice powder was dominated by spore forming bacteria, which was expected considering the low water activity of dried spices.

In general, food products with microbial counts below 6.0 log<sub>10</sub>CFU/g are recommended for safe consumption; counts between 6.0 and 7.0 log<sub>10</sub>CFU/g are considered acceptable, while those exceeding 7.0 log<sub>10</sub>CFU/g are deemed unsatisfactory and potentially unsafe (Kokkinakis et al., 2020). A TVC result of 3.4 could therefore be regarded as being unquestionably within the safe ranges. Meanwhile, a safety limit of 4.0 log<sub>10</sub>CFU/g is recommended for sporeformers, yeast, and molds (Berg et al., 2017).



Five-spice powder was once again found to be within acceptable safety limits across all three aforementioned tests. Furthermore, the test for the common bacterium *Clostridium perfringens*, which is commonly found in many herbs and spices, produced a low reading of <0.3 MPN/g, indicating its safety for consumption.

Compared to the results from other spices (Annex Table B1), five-spice had a relatively low microbial count, suggesting that its microbiological quality was on par with black peppercorns, star anise and Italian herbs. This may be attributed to the antimicrobial activity of cloves and cinnamon, which were present in the five-spice mix. Cloves and cinnamon contain eugenol and cinnamaldehyde respectively, which can act as antimicrobials that inhibit the bacterial and fungal growth (Adams et al., 2024).

### **3.2. Fermented foods (Natto)**

Natto demonstrated significantly low and safe levels in yeast, mold, and lactic acid bacteria (LAB) tests. However, it showed relatively high counts in both the TVC and spore former tests. This was expected because natto is intentionally inoculated with a spore forming bacterium—*Bacillus subtilis* to act as a fermentative agent. Most typically, fermented foods are intentionally inoculated with microbes, and thus a high count is not necessarily indicative of spoilage. Thus, the standard upper threshold for microbial spoilage (7.0 log<sub>10</sub>CFU/g) is not always applicable for fermented products.

Compared to the results from other fermented products (Annex Table B2), natto stands out among the tested fermented food products with the highest TVC reading at >8.8 log<sub>10</sub>CFU/g, significantly surpassing other samples such as kimchi (7.0 log<sub>10</sub>CFU/g), tempeh (7.6 log<sub>10</sub>CFU/g), and yogurt (3.8 log<sub>10</sub>CFU/g), indicating a highly dense microbial population. Additionally, natto showed a relatively high spore former count at 4.8 log<sub>10</sub>CFU/g, further supporting the presence of *Bacillus subtilis* and its spore-forming capability. Conversely, other samples, including kombucha, Yakult, and miso paste, displayed significantly lower TVC and spore former readings, suggesting that natto's distinct microbial profile is characterized by both its robust fermentative bacterial activity and spore formation.

### 3.3. Fresh produce (*Enoki mushroom*)

Enoki mushrooms were subjected to TVC, yeast and mold, LAB, and *Enterobacteriaceae* testing. Enoki yielded a notably high TVC of 8.4 log<sub>10</sub>CFU/g, which was above the typical spoilage threshold level of 7.0 log<sub>10</sub>CFU/g. This could be attributed to the fact that the enoki samples used for testing included the entire fungal structure—encompassing the stipe, base, hyphae, and mycelium at the lower portion. The bottom section may retain soil or dirt residues and is typically trimmed and discarded prior to consumption, which could have contributed to the higher microbial readings observed.

Detected microbes include LAB, spore forming bacteria, yeasts and molds, all present at relatively high levels (around 4.5 to 5.3 log<sub>10</sub>CFU/g). Notably, the *Enterobacteriaceae* test yielded results that were below the detection limit, unexpectedly suggesting their absence. This could be due to *Enterobacteriaceae* strains being outcompeted by other more dominant strains, such as *L. monocytogenes*—a common cause of listeriosis outbreaks originating from enoki mushrooms (Pereira et al., 2023). Additionally, the presence of LAB may have created a low pH environment or bacteriocins such as nisin, that may inhibit the growth of *Enterobacteriaceae*.

Compared to the results from other vegetable and edible fungi products (Annex Table B3), enoki mushrooms and dou miao (pea sprouts) were compared as both are vegetable-based products, making them suitable for assessing differences in microbial profiles. Dou miao showed a TVC reading of more than 9.26 log<sub>10</sub>CFU/g, whereas enoki mushrooms had a TVC reading of 8.4 log<sub>10</sub>CFU/g, indicating a significantly higher microbial load. In contrast to enoki mushrooms—which had lower levels of both mold and yeast (5.3 log<sub>10</sub>CFU/g)—dou miao also displayed noticeably higher levels of both (8.0 log<sub>10</sub>CFU/g). Additionally, LAB were more prevalent in enoki mushrooms (5.8 log<sub>10</sub>CFU/g) than in dou miao (<3.0 log<sub>10</sub>CFU/g). In contrast, *Enterobacteriaceae* was detected at >7.5 log<sub>10</sub>CFU/g in dou miao but remained <1.0 log<sub>10</sub>CFU/g in enoki mushrooms. These noticeable differences may result from dou miao's higher moisture content, delicate tissue structure, and higher potential for spoilage, all of which foster microbial growth. In addition, compared to the relatively better-preserved enoki mushrooms, dou miao showed visible signs of spoilage by the time of microbial investigation, which was potentially caused by poor handling, storage conditions, and time since harvest.

## 4 Conclusions and Recommendations

In conclusion, five-spice powder exhibited outstanding microbial quality, which was most likely attributed to its low water activity and natural antimicrobial properties from various compounds, such as eugenol and cinnamaldehyde. Although natto displayed high TVC and sporeformer counts, it remained safe for consumption as natto production relies on the intentional inoculation of *Bacillus subtilis*—the main fermentative bacteria for natto. Conversely, enoki mushrooms displayed elevated microbial loads, especially in TVC testing, raising concerns over spoilage and potential contamination. It is generally recommended that enoki mushrooms be thoroughly trimmed, washed, and cooked before consumption. Storage conditions could also be optimized to reduce microbial proliferation. Moving forward, greater attention should be placed on promoting public education regarding appropriate food preparation techniques. Ongoing microbiological monitoring is recommended, particularly for high-risk products, to ensure greater food safety.

## References

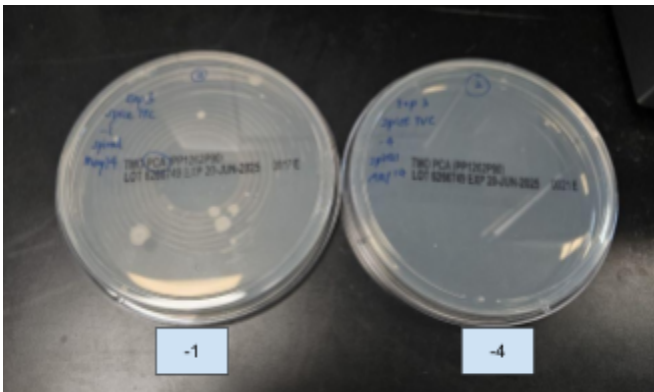
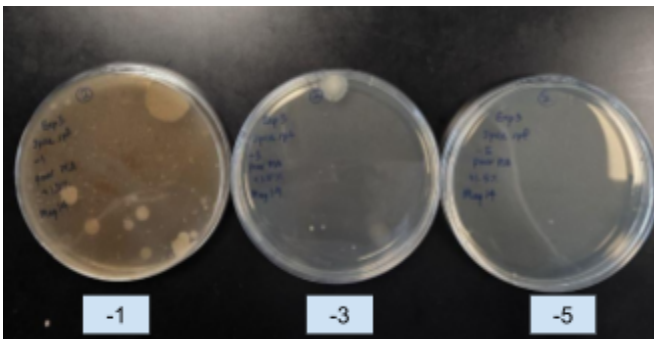
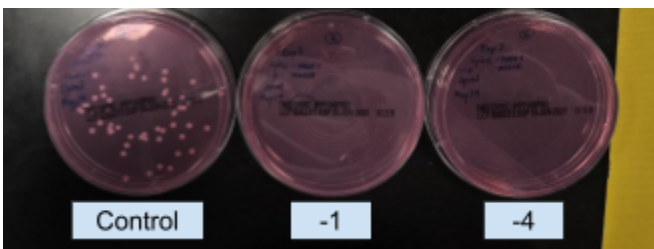
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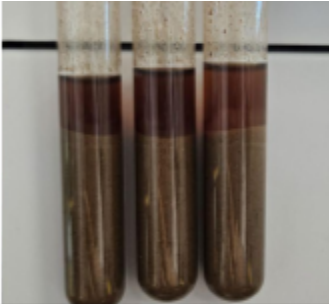
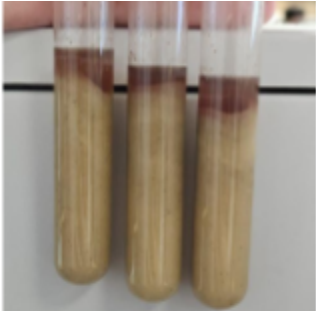
## Annex

**Table A1:** Microbiological quality of selected herbs and spices—five-spice powder

| Group of m.o.         | Medium | Type of plate | Lower detection limit (log <sub>10</sub> CFU /g) | Dilution factor | CFU counted (Prelimin ary) | CFU counted (After confirmati on) | Concentration/ log <sub>10</sub> CFU/g                                                |
|-----------------------|--------|---------------|--------------------------------------------------|-----------------|----------------------------|-----------------------------------|---------------------------------------------------------------------------------------|
| TVC                   | PCA    | Spiral plate  | 2.3                                              | -1              | 12                         | 12                                | 12 x 10 x 20 = 2400<br>log <sub>10</sub> (2.4 x 10 <sup>3</sup> ) = <b><u>3.4</u></b> |
|                       |        |               |                                                  | -4              | <1                         | <1                                |                                                                                       |
| Spf                   | PCA    | Pour plate    | 1.0                                              | -1              | 57                         | 57                                | 57 x 10 x 1 = 570<br>log <sub>10</sub> (5.7 x 10 <sup>2</sup> ) = <b><u>2.8</u></b>   |
|                       |        |               |                                                  | -3              | 7                          | 7                                 |                                                                                       |
|                       |        |               |                                                  | -5              | <1                         | <1                                |                                                                                       |
| Yeast                 | DRBC   | Spiral plate  | 2.3                                              | Control         | 52                         | 54                                | 1 x 10 x 20 = 200<br>log <sub>10</sub> (2 x 10 <sup>2</sup> ) = <b><u>&lt;2.3</u></b> |
|                       |        |               |                                                  | -1              | <1                         | <1                                |                                                                                       |
|                       |        |               |                                                  | -4              | <1                         | <1                                |                                                                                       |
| C. perfrin gens MPN/g | N/A    |               |                                                  |                 | <1                         | <1                                | MPN code: 0-0-0 <b><u>&lt;0.3</u></b>                                                 |

**Table A2:** Images of the tests used to determine the microbiological quality of selected herbs and spices—five-spice powder

| Group of m.o. | Dilution factor | CFU counted (Preliminary) | CFU counted (After confirmation) | Results                                                                                                                                                                                                       |
|---------------|-----------------|---------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TVC</b>    | -1              | 12                        | 12                               |                                                                                                                             |
|               | -4              | <1                        | <1                               |                                                                                                                                                                                                               |
| <b>Spf</b>    | -1              | 57                        | 57                               |  <p>*Sample for -1 dilution is slightly brown not due to microbial growth, but the colour of the five-spice suspension</p> |
|               | -3              | 7                         | 7                                |                                                                                                                                                                                                               |
|               | -5              | <1                        | <1                               |                                                                                                                                                                                                               |
| <b>Yeast</b>  | Control         | 52                        | 54                               |                                                                                                                           |
|               | -1              | <1                        | <1                               |                                                                                                                                                                                                               |
|               | -4              | <1                        | <1                               |                                                                                                                                                                                                               |

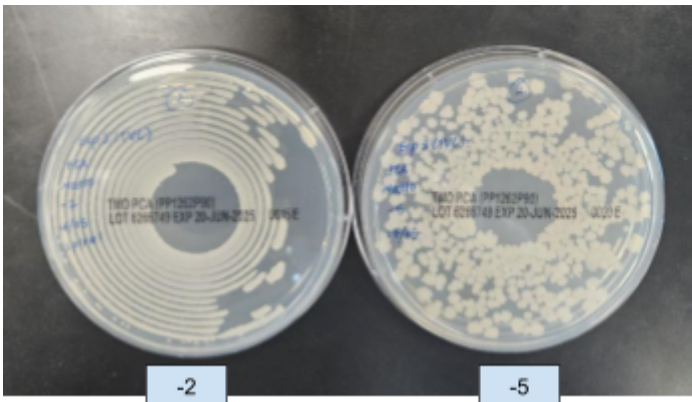
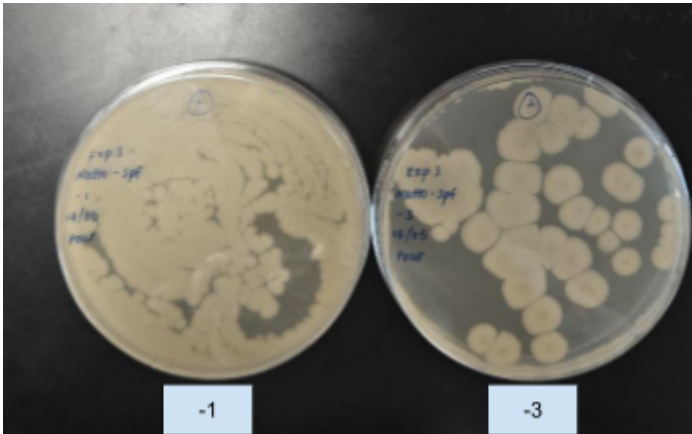
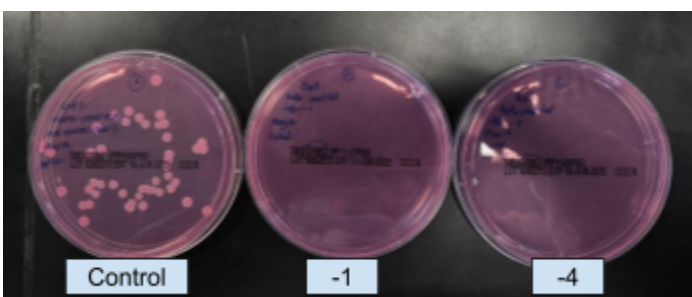
|                                          |                                                         |    |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|---------------------------------------------------------|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C.<br/>perfrin<br/>gens<br/>MPN/g</b> | -1a, -1b,<br>-1c, -2a,<br>-2b, -2c,<br>-3a, -3b,<br>-3c | <1 | <1 | <div data-bbox="907 205 1273 1209">  <div data-bbox="911 512 1273 560"> -1a   -1b   -1c </div>  <div data-bbox="911 877 1273 926"> -2a   -2b   -2c </div>  <div data-bbox="911 1163 1273 1209"> -3a   -3b   -3c </div> </div> <p data-bbox="781 1220 1403 1306">*All samples showed no stormy fermentation, no gas formation or coagulation (brown sediments at the bottom were found to be the five-spice powder)</p> |
|------------------------------------------|---------------------------------------------------------|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

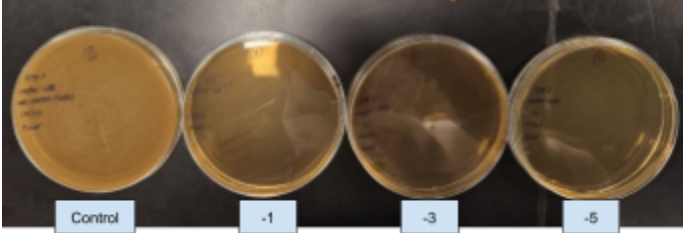
**Table A3:** Microbiological quality of selected fermented product—natto

| Group of m.o. | Medium | Type of plate | Lower detection limit ( $\log_{10}$ CFU/g) | Dilution factor | CFU counted (Preliminary) | CFU counted (After confirmation) | Concentration/<br>$\log_{10}$ CFU/g                                                                 |
|---------------|--------|---------------|--------------------------------------------|-----------------|---------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>TVC</b>    | PCA    | Spiral plate  | 2.3                                        | -2              | >300                      | >300                             | $300 \times 10^5 \times 20 = 600000000$<br>$\log_{10}(6 \times 10^8) = \underline{\underline{8.8}}$ |
|               |        |               |                                            | -5              | >300                      | >300                             |                                                                                                     |
| <b>Spf</b>    | PCA    | Pour plate    | 1.0                                        | -1              | >300 (overgrown)          | >300 (overgrown)                 | $61 \times 1000 \times 1 = 61000$<br>$\log_{10}(6.1 \times 10^4) = \underline{\underline{4.8}}$     |
|               |        |               |                                            | -3              | 61                        | 61                               |                                                                                                     |
| <b>Yeast</b>  | DRBC   | Spiral plate  | 2.3                                        | Control         | 46                        | 50                               | $1 \times 10^1 \times 20 = 200$<br>$\log_{10}(2 \times 10^2) = \underline{\underline{2.3}}$         |
|               |        |               |                                            | -1              | <1                        | <1                               |                                                                                                     |
|               |        |               |                                            | -4              | <1                        | <1                               |                                                                                                     |
| <b>LAB</b>    | MRSA   | Pour plate    | 1.0                                        | Control         | >300 (overgrown)          | >300 (overgrown)                 | $1 \times 10 = 10$<br>$\log_{10}(10) = \underline{\underline{1.0}}$                                 |
|               |        |               |                                            | -1              | <1                        | <1                               |                                                                                                     |
|               |        |               |                                            | -3              | <1                        | <1                               |                                                                                                     |
|               |        |               |                                            | -5              | <1                        | <1                               |                                                                                                     |



**Table A4:** Images of the tests used to determine the microbiological quality of selected fermented product—natto

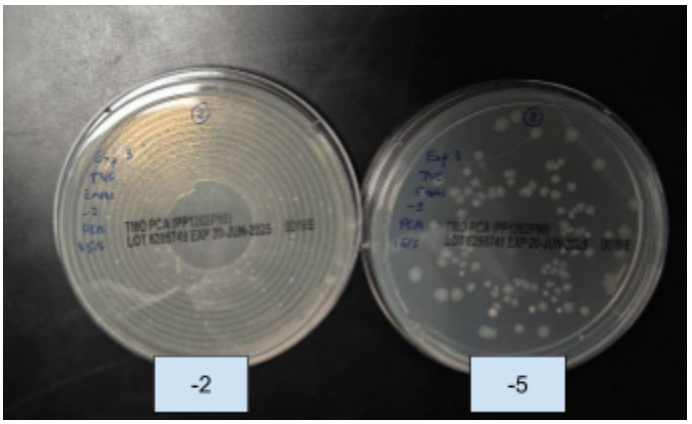
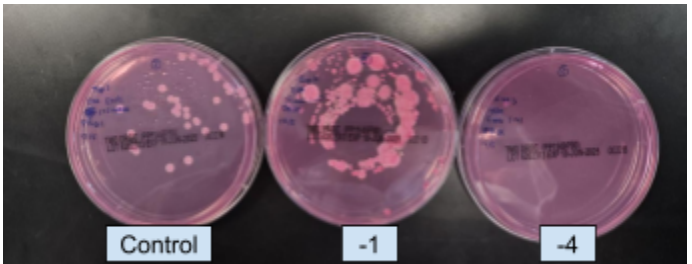
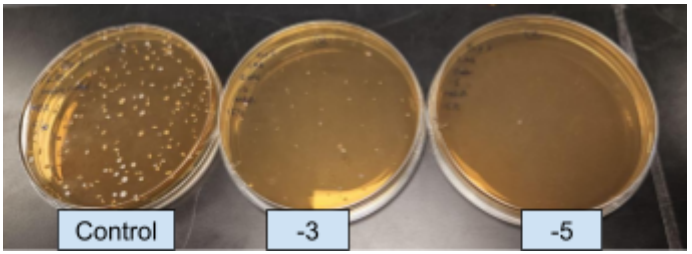
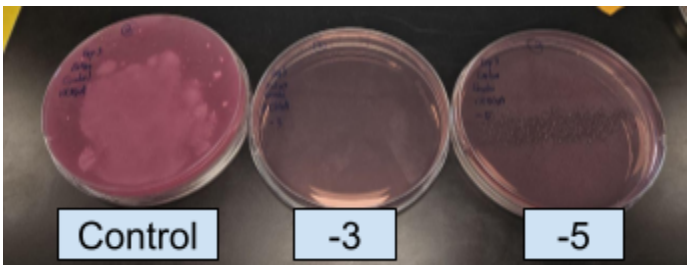
| Group of m.o. | Dilution factor | CFU counted (Preliminary) | CFU counted (After confirmation) | Results                                                                              |
|---------------|-----------------|---------------------------|----------------------------------|--------------------------------------------------------------------------------------|
| <b>TVC</b>    | -2              | >300                      | >300                             |    |
|               | -5              | >300                      | >300                             |                                                                                      |
| <b>Spf</b>    | -1              | >300 (overgrown)          | >300 (overgrown)                 |   |
|               | -3              | 61                        | 61                               |                                                                                      |
| <b>Yeast</b>  | Control         | 46                        | 50                               |  |
|               | -1              | <1                        | <1                               |                                                                                      |
|               | -4              | <1                        | <1                               |                                                                                      |

|     |         |                     |                     |                                                                                    |
|-----|---------|---------------------|---------------------|------------------------------------------------------------------------------------|
| LAB | Control | >300<br>(overgrown) | >300<br>(overgrown) |  |
|     | -1      | <1                  | <1                  |                                                                                    |
|     | -3      | <1                  | <1                  |                                                                                    |
|     | -5      | <1                  | <1                  |                                                                                    |

**Table A5:** Microbiological quality of selected edible fungi—enoki mushroom

| Group of m.o.                    | Medium | Type of plate | Lower detection limit ( $\log_{10}$ CFU/g) | Dilution factor | CFU counted (Preliminary) | CFU counted (After confirmation) | Concentration/<br>$\log_{10}$ CFU/g                                                                     |
|----------------------------------|--------|---------------|--------------------------------------------|-----------------|---------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>TVC</b>                       | PCA    | Spiral plate  | 2.3                                        | -2              | >300                      | >300                             | $115 \times 10^5 \times 20 = 230,000,000$<br>$\log_{10}(2.3 \times 10^8) = \underline{\underline{8.4}}$ |
|                                  |        |               |                                            | -5              | 115                       | 115                              |                                                                                                         |
| <b>Yeast</b>                     | DRBC   | Spiral plate  | 2.3                                        | Control         | 44                        | 44                               | $1 \times 10^4 \times 20 = 200000$<br>$\log_{10}(2 \times 10^5) = \underline{\underline{5.3}}$          |
|                                  |        |               |                                            | -1              | >300                      | >300                             |                                                                                                         |
|                                  |        |               |                                            | -4              | 1                         | 1                                |                                                                                                         |
| <b>LAB</b>                       | MRSA   | Pour plate    | 1.0                                        | Control         | >300                      | >300                             | $35 \times 10^3 = 35000$<br>$\log_{10}(3.5 \times 10^4) = \underline{\underline{4.5}}$                  |
|                                  |        |               |                                            | -3              | 61                        | $61 \times (4/7) = 35$           |                                                                                                         |
|                                  |        |               |                                            | -5              | <1                        | <1                               |                                                                                                         |
| <b><i>Enterobacteriaceae</i></b> | VRBG   | Pour plate    | 1.0                                        | Control         | >300                      | >300                             | $1 \times 10^3 = 1000$<br>$\log_{10}(10^3) = \underline{\underline{3.0}}$                               |
|                                  |        |               |                                            | -3              | <1                        | <1                               |                                                                                                         |
|                                  |        |               |                                            | -5              | <1                        | <1                               |                                                                                                         |

**Table A6:** Images of the tests used to determine the microbiological quality of selected edible fungi—enoki mushroom

| Group of m.o.                    | Dilution factor | CFU counted (Preliminary) | CFU counted (After confirmation) | Results                                                                              |
|----------------------------------|-----------------|---------------------------|----------------------------------|--------------------------------------------------------------------------------------|
| <b>TVC</b>                       | -2              | >300                      | >300                             |    |
|                                  | -5              | 115                       | 115                              |                                                                                      |
| <b>Yeast</b>                     | Control         | 44                        | 44                               |   |
|                                  | -1              | >300                      | >300                             |                                                                                      |
|                                  | -4              | 1                         | 1                                |                                                                                      |
| <b>LAB</b>                       | Control         | >300                      | >300                             |  |
|                                  | -3              | 61                        | 61 x (4/7) = 35                  |                                                                                      |
|                                  | -5              | <1                        | <1                               |                                                                                      |
| <b><i>Enterobacteriaceae</i></b> | Control         | >300                      | >300                             |  |
|                                  | -3              | <1                        | <1                               |                                                                                      |
|                                  | -5              | <1                        | <1                               |                                                                                      |

**Table B1:** Comparison of microbiological quality of selected herbs and spices across all groups

| Category 1: Herbs and Spices |                                                      |                                                                                  |                                                 |                                                |                                                                              |
|------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------|
| Product                      | Total Viable Count (TVC)/<br>log <sub>10</sub> CFU/g | Spore-forming bacteria (spf),<br>spore concentration/<br>log <sub>10</sub> CFU/g | Yeast concentration/<br>log <sub>10</sub> CFU/g | Mold concentration/<br>log <sub>10</sub> CFU/g | <i>C. perfringens</i> concentration,<br>Most Probable Number (MPN)/<br>MPN/g |
| Turmeric Powder              | 6.5                                                  | 6.7                                                                              | 2.8                                             | <2.3                                           | 2.9                                                                          |
| Five Spice Powder            | 3.4                                                  | 2.8                                                                              | <2.3                                            | <2.3                                           | <0.3                                                                         |
| Rosemary                     | 4.3                                                  | 2.4                                                                              | <2.3                                            | <2.3                                           | <0.3                                                                         |
| Cinnamon Powder              | 4.0                                                  | 4.5                                                                              | <2.3                                            | <2.3                                           | 0.36                                                                         |
| Jasmine Tea                  | 3.3                                                  | 2.6                                                                              | <2.3                                            | 3.3                                            | 4.3                                                                          |
| Italian Herbs                | <2.3                                                 | 2.6                                                                              | <2.3                                            | <2.3                                           | 0.92                                                                         |
| Black Peppercorn             | <2.3                                                 | <1.0                                                                             | <2.3                                            | <2.3                                           | <0.3                                                                         |
| Star Anise                   | <2.3                                                 | <1.0                                                                             | <2.3                                            | <2.3                                           | <0.3                                                                         |
| Dried Chrysanthemum Flower   | 4.6                                                  | 4.4                                                                              | <2.3                                            | <2.3                                           | 0.92                                                                         |
| Thyme                        | 3.8                                                  | 5.1                                                                              | <2.3                                            | <2.3                                           | <0.3                                                                         |

**Table B2:** Comparison of microbiological quality of selected fermented products across all groups

| Category 2: Fermented Products   |                                                      |                                                                                  |                                                 |                                              |                                                                      |
|----------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------|
| Product                          | Total Viable Count (TVC)/<br>log <sub>10</sub> CFU/g | Spore-forming bacteria (spf),<br>spore concentration/<br>log <sub>10</sub> CFU/g | Yeast concentration/<br>log <sub>10</sub> CFU/g | Mold concentration / log <sub>10</sub> CFU/g | Lactic Acid Bacteria (LAB) concentration/<br>log <sub>10</sub> CFU/g |
| Kombucha                         | <2.3                                                 | <2.3                                                                             | 2.8                                             | <2.3                                         | <2.0                                                                 |
| Natto                            | >8.8                                                 | 4.8                                                                              | <2.3                                            | <2.3                                         | <1.0                                                                 |
| Kimchi                           | 7.0                                                  | 2.8                                                                              | 3.7                                             | 3.7                                          | >7.5                                                                 |
| Yogurt                           | 3.8                                                  | 5.2                                                                              | 5.3                                             | 5.3                                          | 2.6                                                                  |
| Tempeh                           | 7.6                                                  | 3.3                                                                              | 6.9                                             | 7.2                                          | >7.5                                                                 |
| Yakult                           | <2.3                                                 | <2.3                                                                             | <2.3                                            | <2.3                                         | >7.5                                                                 |
| Fermented Soybean Paste          | 5.7                                                  | >3.5                                                                             | <2.3                                            | <2.3                                         | <1.0                                                                 |
| Honey Mead                       | 3.6                                                  | <2.3                                                                             | 2.4                                             | <2.3                                         | <2.3                                                                 |
| Miso Paste                       | 3.3                                                  | 2.8                                                                              | <2.3                                            | <2.3                                         | 1.5                                                                  |
| Belacan (Fermented Shrimp Paste) | 3.8                                                  | 3.1                                                                              | <2.3                                            | <2.3                                         | <1.0                                                                 |

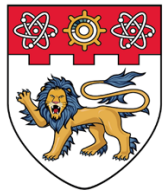
**Table B3:** Comparison of microbiological quality of selected fungi, vegetables and fruits across all groups

| Category 3A: Fungi                                                      |                                 |                                                    |                                                   |                                                  |                                                                        |
|-------------------------------------------------------------------------|---------------------------------|----------------------------------------------------|---------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------|
| Product                                                                 | TVC/<br>log <sub>10</sub> CFU/g | Yeast<br>concentration/<br>log <sub>10</sub> CFU/g | Mold<br>concentration/<br>log <sub>10</sub> CFU/g | LAB<br>concentration/<br>log <sub>10</sub> CFU/g | <i>Enterobacteriaceae</i><br>concentration/<br>log <sub>10</sub> CFU/g |
| Enoki mushroom                                                          | 8.4                             | 5.3                                                | 5.3                                               | 5.8                                              | <1.0                                                                   |
| Category 3B: Vegetables                                                 |                                 |                                                    |                                                   |                                                  |                                                                        |
| Product                                                                 | TVC/<br>log <sub>10</sub> CFU/g | Yeast<br>concentration/<br>log <sub>10</sub> CFU/g | Mold<br>concentration/<br>log <sub>10</sub> CFU/g | LAB<br>concentration/<br>log <sub>10</sub> CFU/g | <i>Enterobacteriaceae</i><br>concentration/<br>log <sub>10</sub> CFU/g |
| Dou miao (Pea sprouts, noticeably spoilt by the time of investigation)  | >9.3                            | 8.2                                                | 8.2                                               | <3.0                                             | >7.5                                                                   |
| Category 3C: Fruits                                                     |                                 |                                                    |                                                   |                                                  |                                                                        |
| Product                                                                 | TVC/<br>log <sub>10</sub> CFU/g | Yeast<br>concentration/<br>log <sub>10</sub> CFU/g | Mold<br>concentration/<br>log <sub>10</sub> CFU/g | LAB<br>concentration/<br>log <sub>10</sub> CFU/g | <i>Enterobacteriaceae</i><br>concentration/<br>log <sub>10</sub> CFU/g |
| Fruit cup (Pre-cut fruit from one day before the time of investigation) | 5.4                             | 5.3                                                | 5.3                                               | 4.9                                              | >3.5                                                                   |

**Table B4:** Comparison of microbiological quality of selected raw meat/ fish, ice cream and dairy, bread or pastry products across all groups

| Category 4A: Raw Meat/ Fish                  |                                     |                                                                    |                                                            |                                                           |                                                          |                                                                                   |                                                                                    |                                                                                       |
|----------------------------------------------|-------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Product                                      | TVC/<br>log <sub>10</sub> CFU/<br>g | Spf,<br>spore<br>concentra<br>tion/<br>log <sub>10</sub> CFU/<br>g | Yeast<br>concentr<br>ation/<br>log <sub>10</sub> CFU/<br>g | Mold<br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | LAB<br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | <i>Enterobacteriac</i><br><i>eae</i><br>concentration/<br>log <sub>10</sub> CFU/g | <i>Pseudo</i><br><i>monas</i><br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | <i>Bacillus</i><br><i>Cereus</i><br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g |
| Raw Minced Chicken                           | >7.8                                | Not done                                                           | 4.1                                                        | Not done                                                  | Not done                                                 | 2.6                                                                               | >4.8                                                                               | Not done                                                                              |
| Ready-To-Eat Shrimp                          | 2.8                                 | Not done                                                           | Not done                                                   | Not done                                                  | <1.0                                                     | <1.0                                                                              | 3.3                                                                                | Not done                                                                              |
| Raw Squid                                    | 4.7                                 | Not done                                                           | Not done                                                   | Not done                                                  | <1.0                                                     | <1.0                                                                              | <2.3                                                                               | Not done                                                                              |
| Raw Salmon (Non-sashimi grade)               | 6.5                                 | Not done                                                           | Not done                                                   | Not done                                                  | <1.0                                                     | >3.5                                                                              | 5.6                                                                                | Not done                                                                              |
| Category 4B: Water, milk, ice cream or dairy |                                     |                                                                    |                                                            |                                                           |                                                          |                                                                                   |                                                                                    |                                                                                       |
| Product                                      | TVC/<br>log <sub>10</sub> CFU/<br>g | Spf,<br>spore<br>concentra<br>tion/<br>log <sub>10</sub> CFU/<br>g | Yeast<br>concentr<br>ation/<br>log <sub>10</sub> CFU/<br>g | Mold<br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | LAB<br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | <i>Enterobacteriac</i><br><i>eae</i><br>concentration/<br>log <sub>10</sub> CFU/g | <i>Pseudo</i><br><i>monas</i><br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | <i>Bacillus</i><br><i>Cereus</i><br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g |
| Pastry cream                                 | 4.8                                 | <1.0                                                               | Not done                                                   | Not done                                                  | <1.0                                                     | 3.5                                                                               | <1.0                                                                               | <1.0                                                                                  |
| Chocolate sundae                             | 2.8                                 | <1.0                                                               | Not done                                                   | Not done                                                  | <1.0                                                     | <1.0                                                                              | <2.3                                                                               | <1.0                                                                                  |
| Category 4C: Bread or pastry                 |                                     |                                                                    |                                                            |                                                           |                                                          |                                                                                   |                                                                                    |                                                                                       |
| Product                                      | TVC/<br>log <sub>10</sub> CFU/<br>g | Spf,<br>spore<br>concentra<br>tion/<br>log <sub>10</sub> CFU/<br>g | Yeast<br>concentr<br>ation/<br>log <sub>10</sub> CFU/<br>g | Mold<br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | LAB<br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | <i>Enterobacteriac</i><br><i>eae</i><br>concentration/<br>log <sub>10</sub> CFU/g | <i>Pseudo</i><br><i>monas</i><br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g | <i>Bacillus</i><br><i>Cereus</i><br>concentr<br>ation/<br>log <sub>10</sub> CFU<br>/g |
| Kimbap (Rice roll)                           | 8.9                                 | <1.0                                                               | 7.4                                                        | <2.3                                                      | >6.4                                                     | <1.0                                                                              | Not done                                                                           | <2.0                                                                                  |





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# Determining the presence of pathogens in food products

CH5200 - Food Microbiology  
Experiment 4



| Group 2                  |                                     |                              |
|--------------------------|-------------------------------------|------------------------------|
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**Date Submitted** : 23 May 2025

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# 1 General Introduction

Foodborne illness accounts for 600 million illnesses, 420,000 deaths, and the loss of 33 million healthy years of life globally annually (Havelaar et al., 2015). The largest culprits of foodborne disease are foodborne pathogens, making the detection of such pathogens an important field of development. In this experiment, the presence of common foodborne pathogens—*Listeria spp.*, *Salmonella spp.*, and *Campylobacter spp.*—was investigated in common food products, including raw meats and vegetables, to assess the health risks present in a diverse spectrum of food products.

## 2 Materials and Methods

In this experiment, a 25g sample of frozen chicken fillet was investigated for the presence of *Campylobacter spp.* by using various biochemical tests and an immunological assay (Singlepath®). The chicken used was pre-cut and packaged on 26th April, bought on 27th April from a grocer, and stored in a home freezer until it was thawed for testing on 15th May. Additionally, 25g samples of broccoli from a home refrigerator (bought on 10th May) were tested on the same day as the chicken for the presence of *Salmonella spp.*, *Listeria spp.*, and *L. monocytogenes* using a diverse spectrum of qualitative tests such as selective culturing, isolation, and confirmatory tests on various media.

### 3 Results and Discussion

This experiment evaluated the presence of pathogenic microorganisms, specifically *Campylobacter* spp., *Listeria* spp., *L. monocytogenes*, and *Salmonella* spp., using a variety of techniques.

**Table 1:** Presence of pathogens in broccoli and frozen chicken fillet samples

| Product                 | <i>Campylobacter</i> spp. | <i>L. monocytogenes</i> | <i>Listeria</i> spp. other than <i>monocytogenes</i> | <i>Salmonella</i> spp. | <i>Salmonella</i> serology group |
|-------------------------|---------------------------|-------------------------|------------------------------------------------------|------------------------|----------------------------------|
| Chicken fillet (Frozen) | -                         | Not done                | Not done                                             | Not done               | Not done                         |
| Broccoli                | Not done                  | +                       | +                                                    | +                      | Non-specific, Vi                 |

From Table 1, it was concluded that *Campylobacter* spp. was not detected in the frozen chicken fillet sample, whereas *Listeria* spp., *L. monocytogenes*, and *Salmonella* spp. (non-specific and Vi serology) were detected in the broccoli sample.

#### 3.1. Detection of *Campylobacter* spp. in frozen chicken fillet sample

Enriched cultures of the chicken sample were streaked on selective mCCDA and CASA plates to detect the presence of *Campylobacter* spp. After incubation, the plates obtained showed no strong colony differences (Annex Table A1) between the chicken sample and the positive control (red colonies for CASA and gray colonies for mCCDA). Colonies were then tested to be oxidase and catalase positive, which is typical of *Campylobacter*.

However, the Singlepath® immunological assay revealed a negative result for the presence of *Campylobacter* in the chicken sample. Upon microscopic visualization, the colonies from the chicken sample were observed to be rod-shaped with a distinctively different morphology from the positive control (Annex Tables A3 and A4). Hence, the chicken sample was interpreted to not contain any detectable *Campylobacter*.

#### 3.2. Detection of *L. monocytogenes* in broccoli sample

To detect the presence of *Listeria* spp. and *L. monocytogenes*, an enrichment culture from the broccoli sample was plated on selective ALOA plates.

From Annex Figure B1, both the positive control and broccoli sample culture showed the presence of blue-green colonies with an opaque halo around the colonies. This indicated the presence of both *Listeria* spp. and *L. monocytogenes* in the broccoli sample.

### **3.3. Detection of *Salmonella* spp. in broccoli sample**

From Annex Table C1, broccoli samples enriched in MSRV and MKTTn media showed indicators of *Salmonella* spp.—gray-white motility zones with colonies radiating outward from the center of the droplet on MSRV plates, pale blue/ gray, turbid MKTTn solution with precipitate observed at the bottom of the tube without mixing. Upon culturing with BGA, pink colonies were observed on both MSRV and MKTTn samples. Compared to culturing with XLD, the MKTTn sample showed black colonies (positive), while the MSRV sample showed no black colonies (negative).

Three out of the four samples showed indicators of *Salmonella* spp.—MSRV BGA, MKTTn BGA and MKTTn XLD—which are then further tested to confirm the presence/ absence of *Salmonella* spp.

From Annex Table C3, all three samples used showed positive results for the TSI test, indicating the presence of *Salmonella*—the yellow slant and black butt of the TSI agar indicating glucose fermentation, along with gas formation and H<sub>2</sub>S production shown by cracks, bubbles, and blackening of the agar. Similarly, all three samples displayed positive results for the latex agglutination test—red agglutination for MKTTn BGA suggesting Vi serogroup, and grey/ purple agglutination for MSRV BGA and MKTTn XLD suggesting a non-specific *Salmonella* serogroup.

However, for the LDB test, only MSRV BGA and MKTTn XLD were found to be positive—turbid purple media, indicating the presence of *Salmonella*. Additionally, for the UA test, only the MSRV BGA sample resulted in a positive result—unchanged yellow slant, indicating the presence of *Salmonella*. The inconsistent results may be due to differences in colony selection, varying test sensitivity, or metabolic variability and viability across *Salmonella* strains. Hence, the broccoli sample was interpreted to contain *Salmonella* spp.

### **3.4. Pathogenic diversity across different foods**

Table 2 shows the summarized results for the presence of detected pathogens across different food samples (Annex Table D1).

**Table 2:** Summary of detected pathogens found on tested food products.

| Product category        | Remarks                                                             |
|-------------------------|---------------------------------------------------------------------|
| Chicken                 | <i>Campylobacter spp.</i> , <i>Salmonella spp.</i> (C, E or G)      |
| Pork                    | <i>Salmonella spp.</i> (Non specific, B)                            |
| Beef                    | No pathogens detected                                               |
| Vegetable (Broccoli)    | <i>L. monocytogenes</i> , <i>Salmonella spp.</i> (Non specific, Vi) |
| Seafood (Blood cockles) | No <i>Listeria spp.</i> detected; not tested for other pathogens.   |

From the results, *Salmonella spp.* was the most commonly identified pathogen and was detected in raw chicken, pork, and vegetable products. *Salmonella*, a rather common pathogenic bacteria, can infect humans through contaminated food products (e.g., raw meat, vegetables, and dairy products), environmental sources (e.g., water sources), and direct contact with human or animal feces (Huoy et al., 2024). Hence, the presence of *Salmonella* was expected due to the products' freshness and pre-processing conditions. However, *Salmonella* was unexpectedly absent from some raw meat samples (beef, chicken and pork). This may be due to the effect of freezing, which can damage *Salmonella* such that it is unable to sufficiently recover in the 24-hour non-specific enrichment that was done (Digirolamo et al., 1970).

Similarly, *Campylobacter spp.* was unexpectedly absent from some raw chicken samples. This can be attributed to the sensitivity of *Campylobacter* to freezing, chilling or atmospheric air. Such conditions may reduce its viability. It is still possible that *Campylobacter spp.* is present in the samples that tested negative but in a damaged/stressed form, requiring a longer recovery period to be detected (Adams et al., 2024).

*L. monocytogenes* was only detected in broccoli and absent in all raw and processed meats. *L. monocytogenes* is sensitive to low pH environments; it has a narrow pH range (6.0-7.0) for optimal growth. Hence, the post-rigor mortis drop in pH in meat products may selectively suppress the growth of *L. monocytogenes* compared to other microbes (Adams et al., 2024). As broccoli has a higher pH than raw meats, *L. monocytogenes* flourishes in it.

Blood cockles, as filter-feeding bivalves, can accumulate contaminants, including bacteria, viruses, and toxins. However, the absence of *L. monocytogenes* may be due to its low concentration in water bodies or limited uptake by the cockles. Additionally, depuration (i.e., a post-harvest cleansing process of bivalves) may have removed contaminants, including *L. monocytogenes* (Teo, 2023).

## 4 Conclusions and Recommendations

Often so, at least one of the pathogens *Salmonella*, *Campylobacter* and *L. monocytogenes* was detected in every food product tested, suggesting that cooking raw foods is still of utmost importance in food safety. *Salmonella* and *Campylobacter* were found to be less ubiquitous than expected, possibly due to freezing. Additionally, *L. monocytogenes* was found to be less prevalent in meats but present in vegetables. Overall, the methods used for isolation and detection were acceptable albeit time-consuming and potentially less sensitive for frozen foods. Hence, it is recommended that longer enrichment times or other methods be used for frozen samples.

The detection of foodborne pathogens still remains as a crucial step in ensuring food safety and security on a national and global scale. This promotes public health and awareness, preventing foodborne outbreaks and ensuring regulatory compliance in the food industry.

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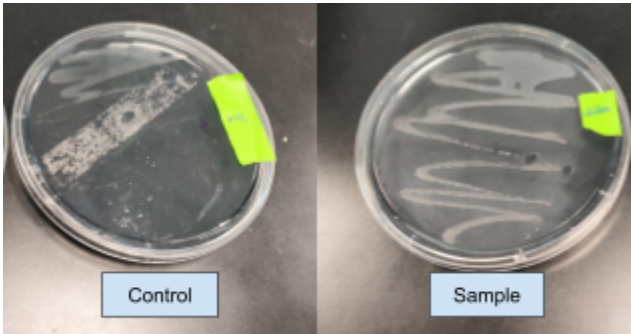
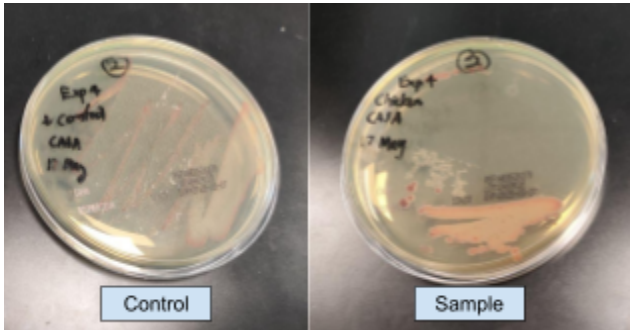
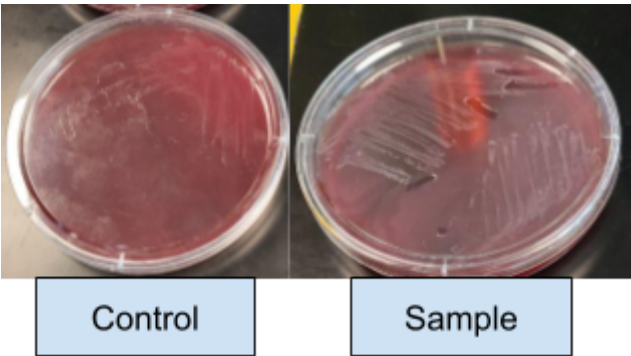
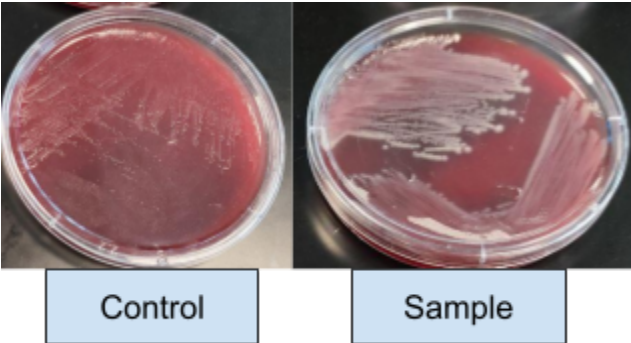


## Annex

**Table A1:** Results of selective enrichment of frozen chicken fillet sample processed using a stomacher (in Modified Charcoal Cefoperazone Desoxycholate Agar [mCCDA] or *Campylobacter* Selective Agar Biomérieux [CASA]); resulting colonies were then isolated on Columbia Agar Base (CAB) plates

| Phase                | Sample                   | Culture medium | Observation                | Result   |
|----------------------|--------------------------|----------------|----------------------------|----------|
| Selective enrichment | Positive control         | mCCDA          | Flat gray colonies present | Positive |
|                      |                          | CASA           | Red colonies present       | Positive |
|                      | Chicken sample           | mCCDA          | Flat gray colonies present | Positive |
|                      |                          | CASA           | Red colonies present       | Positive |
| Final isolates       | mCCDA (Positive control) | CAB            | Flat gray colonies present | N/A      |
|                      | CASA (Positive control)  | CAB            | Flat gray colonies present | N/A      |
|                      | mCCDA (Chicken sample)   | CAB            | Flat gray colonies present | N/A      |
|                      | CASA (Chicken sample)    | CAB            | Flat gray colonies present | N/A      |

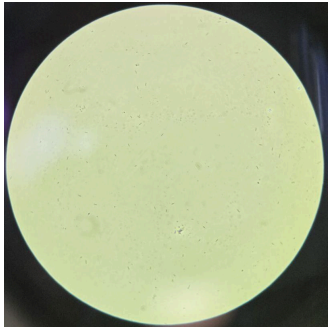
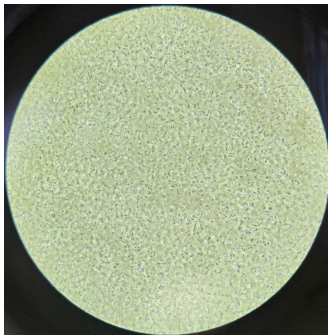
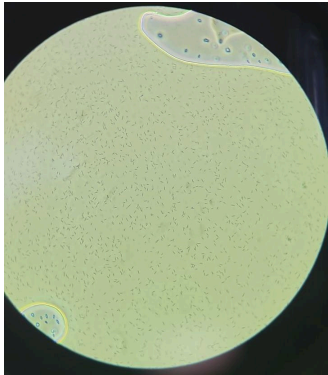
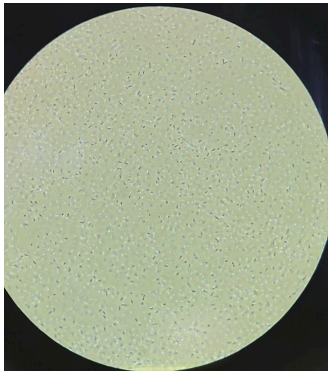
**Table A2:** Images of selective enrichment results from Table A1

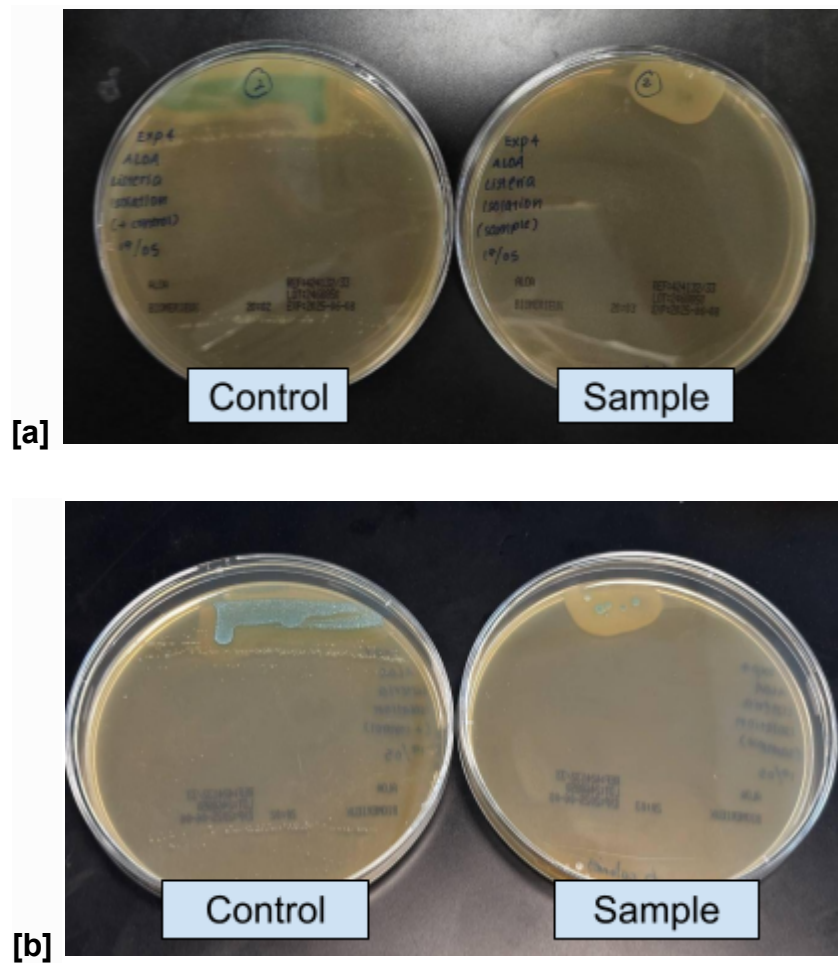
| Phase                | Culture medium | Results                                                                                                                   |  |
|----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------|--|
| Selective enrichment | mCCDA          |  <div>Control</div> <div>Sample</div>   |  |
|                      | CASA           |  <div>Control</div> <div>Sample</div>  |  |
| Final isolates       | mCCDA<br>CAB   |  <div>Control</div> <div>Sample</div> |  |
|                      | CASA<br>CAB    |  <div>Control</div> <div>Sample</div> |  |

**Table A3:** Bacterial identification results of positive controls and chicken samples cultured on mCCDA and CASA media, including oxidase and catalase tests, assessment of cell shape and motility pattern, and Singlepath® testing

| Sample/<br>Test                         | Oxidase  | Catalase | Shape | Motility                             | Singlepath® | Result   |
|-----------------------------------------|----------|----------|-------|--------------------------------------|-------------|----------|
| <b>mCCDA<br/>(Positive<br/>control)</b> | Positive | Positive | Rod   | Motile without<br>corkscrew<br>shape | Positive    | Positive |
| <b>CASA<br/>(Positive<br/>control)</b>  | Positive | Positive | Rod   | Motile without<br>corkscrew<br>shape | Positive    | Positive |
| <b>mCCDA<br/>(Chicken<br/>sample)</b>   | Positive | Positive | Rod   | Motile without<br>corkscrew<br>shape | Negative    | Negative |
| <b>CASA<br/>(Chicken<br/>sample)</b>    | Positive | Positive | Rod   | Motile without<br>corkscrew<br>shape | Negative    | Negative |

**Table A4:** Images of confirmatory test results from Table A3

| Sample/ Test                | Microscopy                                                                          | Singlepath®                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| mCCDA<br>(Positive control) |    |                                        |
| mCCDA<br>(Chicken sample)   |   | <p><b>[control]:</b> Two lines observed (<b>Positive</b>)</p> <p><b>[sample]:</b> One line observed (<b>Negative</b>)</p> |
| CASA<br>(Positive control)  |  |                                      |
| CASA<br>(Chicken sample)    |  | <p><b>[control]:</b> Two lines observed (<b>Positive</b>)</p> <p><b>[sample]:</b> One line observed (<b>Negative</b>)</p> |

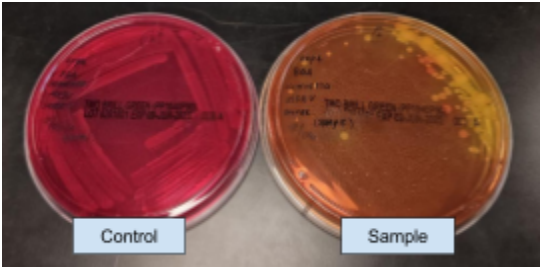
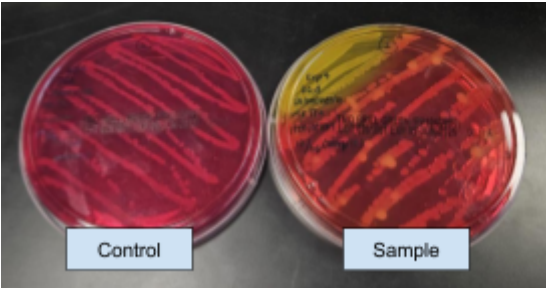
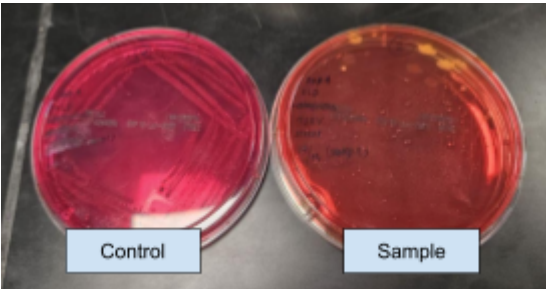
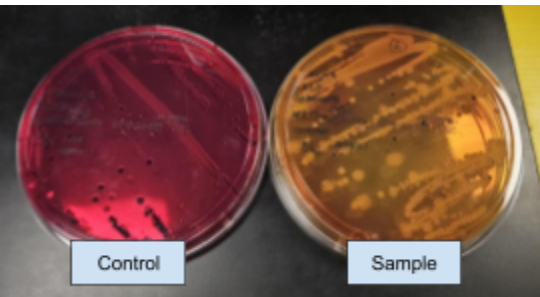


**Figure B1:** [a] Bottom view of Agar *Listeria* Ottaviani Agosti (ALOA) plates, [right] Top view of ALOA plates (clearer view of blue-green colonies and surrounding halo)

**Table C1:** Results of selective enrichment of broccoli sample processed using a stomacher (in Modified Semi-solid Rappaport-Vassiliadis [MSRV] or Muller-Kauffmann tetrathionate-novobiocin broth [MKTTn]); resulting colonies were then isolated on culture media (Brilliant Green Agar [BGA] or Xylose Lysine Desoxycholate [XLD])

| Sample                  | Sample used for isolation | Observation                                                                                          | Culture medium | Observation                        | Result          |
|-------------------------|---------------------------|------------------------------------------------------------------------------------------------------|----------------|------------------------------------|-----------------|
| <b>Positive control</b> | <b>MSRV</b>               | Gray-white motility zones, with colonies radiating outward from the center of the droplet            | <b>BGA</b>     | Pink colonies on cerise medium     | <b>Positive</b> |
|                         | <b>MKTTn</b>              | Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing | <b>BGA</b>     | Pink colonies on cerise medium     | <b>Positive</b> |
|                         | <b>MSRV</b>               | Gray-white motility zones, with colonies radiating outward from the center of the droplet            | <b>XLD</b>     | No black colonies on cerise medium | <b>Positive</b> |
|                         | <b>MKTTn</b>              | Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing | <b>XLD</b>     | Black colonies on cerise medium    | <b>Positive</b> |
| <b>Broccoli sample</b>  | <b>MSRV</b>               | Gray-white motility zones, with colonies radiating outward from the center of the droplet            | <b>BGA</b>     | Pink colonies on cerise medium     | <b>Positive</b> |
|                         | <b>MKTTn</b>              | Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing | <b>BGA</b>     | Pink colonies on cerise medium     | <b>Positive</b> |
|                         | <b>MSRV</b>               | Gray-white motility zones, with colonies radiating outward from the center of the droplet            | <b>XLD</b>     | No black colonies on cerise medium | <b>Negative</b> |
|                         | <b>MKTTn</b>              | Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing | <b>XLD</b>     | Black colonies on cerise medium    | <b>Positive</b> |

**Table C2:** Images of selective enrichment results from Table C1

| Sample used for isolation | Culture medium | Results                                                                              |
|---------------------------|----------------|--------------------------------------------------------------------------------------|
| MSRV                      | BGA            |    |
| MKTTn                     | BGA            |    |
| MSRV                      | XLD            |  |
| MKTTn                     | XLD            |  |

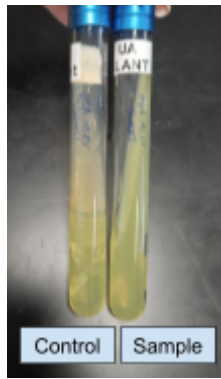
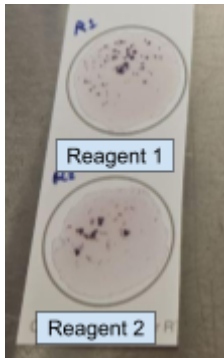
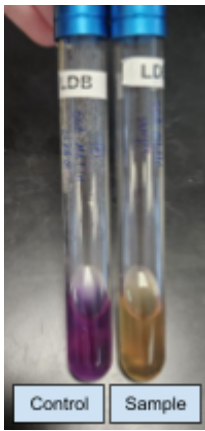
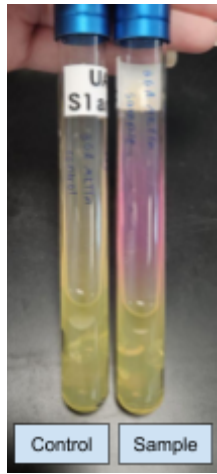
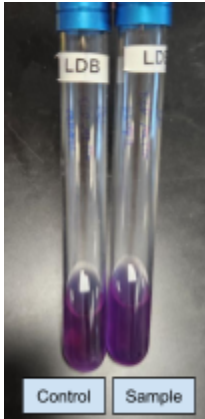
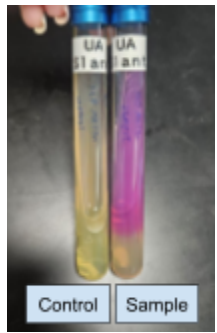


**Table C3:** Results of the confirmation tests using positive colonies from Table 1—MSRV BGA, MKTTn BGA and MKTTn XLD

| Sample/ test     |           | Triple Sugar Iron Agar (TSI)                         | Lysine Decarboxylase Broth (LDB)    | Urea Agar (UA)                 | Latex agglutination                                                              |
|------------------|-----------|------------------------------------------------------|-------------------------------------|--------------------------------|----------------------------------------------------------------------------------|
|                  |           | Observation                                          | Observation                         | Observation                    | Observation                                                                      |
| Positive control | MSRV BGA  | Red slant and black butt <b>(Positive)</b>           | Purple and turbid <b>(Positive)</b> | Yellow slant <b>(Positive)</b> | Not done                                                                         |
|                  | MKTTn BGA | Red slant and black butt <b>(Positive)</b>           | Purple and turbid <b>(Positive)</b> | Yellow slant <b>(Positive)</b> | Not done                                                                         |
|                  | MKTTn XLD | Slight yellow slant and black butt <b>(Positive)</b> | Purple and turbid <b>(Positive)</b> | Yellow slant <b>(Positive)</b> | Not done                                                                         |
| Broccoli sample  | MSRV BGA  | Yellow slant and black butt <b>(Positive)</b>        | Purple and turbid <b>(Positive)</b> | Yellow slant <b>(Positive)</b> | R1: Grey purple agglutination<br>R2: Grey purple agglutination <b>(Positive)</b> |
|                  | MKTTn BGA | Yellow slant and black butt <b>(Positive)</b>        | Yellow and turbid <b>(Negative)</b> | Purple slant <b>(Negative)</b> | R1: Grey purple agglutination<br>R2: Red purple agglutination <b>(Positive)</b>  |
|                  | MKTTn XLD | Slight yellow slant and black butt <b>(Positive)</b> | Purple and turbid <b>(Positive)</b> | Purple slant <b>(Negative)</b> | R1: Grey purple agglutination<br>R2: Grey purple agglutination <b>(Positive)</b> |



**Table C4:** Images of confirmation test results from Table C3

| Sample used and culture medium/<br>Confirmation test results | Triple Sugar Iron Agar (TSI)                                                                          | Lysine Decarboxylase Broth (LDB)                                                                      | Urea Agar (UA)                                                                                         | Latex Agglutination                                                                                             |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| MSRV BGA                                                     | <br>Control Sample   | <br>Control Sample   | <br>Control Sample   | <br>Reagent 1<br>Reagent 2   |
| MKTTn BGA                                                    | <br>Control Sample  | <br>Control Sample  | <br>Control Sample  | <br>Reagent 1<br>Reagent 2  |
| MKTTn XLD                                                    | <br>Control Sample | <br>Control Sample | <br>Control Sample | <br>Reagent 1<br>Reagent 2 |

**Table D1:** Presence of the *Campylobacter* spp., *Listeria* spp., and *Salmonella* spp. in common food products.

| Category 1: Chicken products |                           |                         |                                               |                        |                                  |
|------------------------------|---------------------------|-------------------------|-----------------------------------------------|------------------------|----------------------------------|
| Product                      | <i>Campylobacter</i> spp. | <i>L. monocytogenes</i> | <i>Listeria</i> spp. other than monocytogenes | <i>Salmonella</i> spp. | <i>Salmonella</i> serology group |
| Chicken minced               | -                         | -                       | +                                             | +                      | E or G                           |
| Chicken liver                | +                         | Not done                | Not done                                      | Not done               | Not done                         |
| Chicken fillet (Frozen)      | -                         | Not done                | Not done                                      | Not done               | Not done                         |
| Chicken thigh                | +                         | -                       | +                                             | -                      | -                                |
| Chicken heart                | Not done                  | Not done                | Not done                                      | -                      | -                                |
| Chicken wing                 | -                         | -                       | +                                             | +                      | C, E or G                        |
| Chicken egg                  | -                         | Not done                | Not done                                      | Not done               | -                                |
| Category 2: Pork products    |                           |                         |                                               |                        |                                  |
| Product                      | <i>Campylobacter</i> spp. | <i>L. monocytogenes</i> | <i>Listeria</i> spp. other than monocytogenes | <i>Salmonella</i> spp. | <i>Salmonella</i> serology group |
| Pork minced 1                | -                         | -                       | +                                             | +                      | B                                |
| Pork minced 2                | Not done                  | Not done                | Not done                                      | -                      | -                                |
| Pork liver                   | -                         | -                       | -                                             | +                      | Non specific                     |
| Pig intestine                | -                         | -                       | -                                             | -                      | -                                |
| Picnic ham                   | Not done                  | -                       | -                                             | Not done               | Not done                         |
| Category 3: Beef products    |                           |                         |                                               |                        |                                  |
| Product                      | <i>Campylobacter</i> spp. | <i>L. monocytogenes</i> | <i>Listeria</i> spp. other than monocytogenes | <i>Salmonella</i> spp. | <i>Salmonella</i> serology group |
| Beef minced                  | -                         | -                       | -                                             | -                      | -                                |

| Category 4: Others |                                     |                                   |                                                            |                                  |                                        |
|--------------------|-------------------------------------|-----------------------------------|------------------------------------------------------------|----------------------------------|----------------------------------------|
| Product            | <i>Campylobacter</i><br><i>spp.</i> | <i>L.</i><br><i>monocytogenes</i> | <i>Listeria</i> <i>spp.</i> other<br>than<br>monocytogenes | <i>Salmonella</i><br><i>spp.</i> | <i>Salmonella</i><br>serology<br>group |
| Broccoli           | Not done                            | +                                 | +                                                          | +                                | Vi, Non<br>specific                    |
| Blood<br>cockles   | Not done                            | -                                 | -                                                          | Not done                         | Not done                               |